MouRa - Agri AI

MAT 111 ELEMENTARY MATHEMATICS (1+1)

LECTURE NOTES

B.Sc. (Hons.)(Agriculture)

Course Teacher

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MAT 111 ELEMENTARY MATHEMATICS (1+1)

Objective:

To understand and apply fundamental concepts of mathematics applicable in biology and to acquire about theoretical concepts of Algebra, Geometry, Calculus and Mathematical Modeling.

Theory

Unit - I

Algebra: Permutation and Combination -meaning of nPr and nCr (simple problems). Matrices- Definition of Matrices, Addition, Subtraction, Multiplication, Transpose and Inverse up to 3rd order by adjoint method, Properties of determinants up to 3rd order and their evaluation**.**

Unit - II

Analytical Geometry: Distance formula, section formula (internal and external division), Change of axes (only origin changed), Equation of co-ordinate axes, Equation of lines parallel to axes, Slope-intercept form of equation of line, Slope-point form of equation of line, Two point form of equation of line, Intercept form of equation of line, Normal form of equation of line, General form of equation of line, Point of intersection of two straight lines, Angles between two straight lines, Parallel lines, Perpendicular lines.

Equation of circle whose centre and radius is known, General equation of a circle, Equation of circle passing through three given points, Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2,y_2).

Unit - III

Differential Calculus: Definition of function, limit and continuity, Simple problems on limit, Simple problems on continuity, Differentiation of x^n , e^x , sin x &cos x from first principle, Derivatives of sum, difference, product and quotient of two functions, Differentiation of functions of functions (Simple problem based on it), Logarithmic differentiation (Simple problem based on it), Differentiation by substitution method and simple problems based on it, Differentiation of Inverse Trigonometric functions. Partial differentiation with first and second order -Maxima and Minima of the functions of the form y = f (x) and y = $f(x_1,x_2)$ (Simple problems based on it).

Unit –IV

Integral Calculus: Integration of simple functions, Integration of Product of two functions, Integration by substitution method, Definite Integral (simple problems based on it), Area under simple well-known curves (simple problems based on it).

Unit-V

Mathematical Models: Agricultural systems - Mathematical models - classification of mathematical models- Fitting of Linear, quadratic and exponential models to experimental data.

Practical

Simple problems in Permutation and Combination -meaning of nPr and nCr Problems in Algebra of matrices , Transpose and Inverse up to 3rd order by adjoint method, evaluation of determinants up to 3rd order**.** Problems in Straight lines using distance formula, section formula (internal and external division), Change of axes (only origin changed)- Equation of co-ordinate axes- Equation of lines parallel to axes. Problems in equation of a line in : Slope-intercept form, Slope-point form, two point forms, Intercept form, Normal form , General form, Point of intersection of two straight lines. Problems in Angles between two straight lines, Parallel lines, Perpendicular lines. Problems in Equation of circle whose centre and radius is known, General equation of a circle, Equation of circle passing through three given points, Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2,y_2). Simple problems in limit and continuity. Problems in differentiation of x^n , e^x , sin x &cos x, derivatives of sum, difference, product and quotient of two functions. Simple problem based on differentiation of functions of functions and Logarithmic differentiation. Simple problems based on differentiation by substitution method. Problems in partial differentiation and Maxima and Minima of the functions of the form y=f (x)and $y=f(x_1,x_2)$. Problems in integration of simple functions and product of two functions- Definite Integral. Integration by substitution method-Problems in Area under simple well-known curves. Problems in fitting linear, quadratic and Exponential models to experimental data.

Theory Schedule:

1. Permutation and Combination -meaning of nPr and nCr (Simple Problems) .
2. Matrices- Definition of Matrices- Types of Matrices- Addition, Subtraction, Multiplication, Transpose
3. Determinants-Properties of determinants -up to 3rd order evaluation and inverse up to 3rd order by adjoint method.
4. Straight lines - Distance formula-section formula (internal and external division) - Change of axes (only origin changed) - Equation of co-ordinate axes- Equation of lines parallel to axes.
5. Forms of equation of Line-Slope-intercept form -Slope one point form - Two point form -Intercept form.
6. Normal form of equation of line- General form of equation of line- Point of intersection of two straight lines.
7. Angles between two straight lines- Parallel lines- Perpendicular lines- Angle of bisectors between two lines.
8. Circle-Equation of circle whose centre and radius is known- General equation of a circle- Equation of circle passing through three given points- Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2,y_2).

9. **Mid Semester Examination**

10. Differential Calculus - Definition of function, limit and continuity- Simple problems on limit and continuity.

11. Differentiation of x^n , e^x , sin x &cos x from first principle-Derivatives of sum, difference, product and quotient of two functions- Differentiation using functions of function rule (Simple problem based on it)
12. Logarithmic differentiation (Simple problem based on it)- Differentiation by substitution method and simple problems based on it- Differentiation of Inverse Trigonometric functions
13. Maxima and Minima of the functions of the form y=f (x) and y=f(x_1,x_2) (Simple problems based on it).
14. Integral Calculus - Integration of simple functions and Product of two functions- Definite Integral (simple problems based on it)
15. Integration by substitution method- Area under simple well-known curves (simple problems based on it).
16. Agricultural systems - Mathematical models - classification of mathematical models- Linear model.
17. Quadratic and Exponential models- applications of mathematical models in agriculture.

Practical Schedule:

1. Simple problems in Permutation and Combination.
2. Problems in Addition, Subtraction, Multiplication and Transpose of a matrix
3. Problems in determinants and Inverse up to 3rd order by adjoint method.
4. Problems in Straight lines using distance formula, section formula (internal and external division), Change of axes (only origin changed)- Equation of co-ordinate axes- Equation of lines parallel to axes.
5. Problems in Slope-intercept form of equation of line, Slope-point form of equation of line, two point forms of equation of line, Intercept form of equation of line.
6. Problems in Normal form of equation of line, General form of equation of line, Point of intersection of two straight lines.
7. Problems in Angles between two straight lines, Parallel lines, Perpendicular lines, Angle of bisectors between two lines.
8. Problems in Equation of circle whose centre and radius is known, General equation of a circle, Equation of circle passing through three given points, Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2,y_2).
9. Simple problems in limit and continuity.
10. Problems in differentiation of x^n , e^x , sin x &cos x, derivatives of sum, difference, product , quotient of two functions and differentiation of functions of functions.
11. Simple problem based on Logarithmic differentiation and differentiation by substitution method.
12. Problems in Maxima and Minima of the functions of the form y=f (x) and y=f(x_1,x_2)

13. Problems in integration of simple functions and product of two functions using integration by parts- Definite Integral.
14. Integration by substitution method-Problems in Area under simple well-known curves
15. Problems in fitting linear models to experimental data .
16. Problems in fitting Quadratic and Exponential models to experimental data.
17. **Final Practical Examination.**

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4. Duraipandian, 2007, Calculus and Analytical Geometry, Emerald Publishers, Chennai.
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UNIT-I

ALGEBRA

LECTURE SCHEDULE-1

PERMUTATION AND COMBINATION – MEANING OF nP_r and nC_r – SIMPLE PROBLEMS

Permutation (or) Arrangement:

Permutation means arrangement of things. The word arrangement is used, if the order of things is considered.

Meaning of n P_r:

The number of permutations n objects taking r objects at a time is denoted by ${}_n\mathrm{Pr}$. Its value is given by

$$ {}_n\mathrm{Pr} = n(n-1)(n-2),...,(n-r+1) $$

Formula: $$ {}_n\mathrm{Pr} = \frac{n!}{(n-r)!} \qquad (1) $$

P**ermutation** relates to arranging all the elements of a set into some order, or if the set is already ordered, rearranging its elements, a process called permuting.

Example

1. The permutations of three letters a, b, c taken two at a time are

   ab, ba, bc, cb, ac, ca.

(ie) The number of these three letters taken two at a time is denoted by $3P_2 = 6$.

2. Evaluate 11P2

Solution: $$ 11P_2 = \frac{11!}{(11-2)!} = \frac{11!}{9!} = \frac{11\times10\times9!}{9!} = 110 $$

$$ 11P_2 = 11(11-1) = 110 \ (\because \text{Deduction: 2}) $$

Fundamental principle of counting:

(a) Addition Principle:

If there are two jobs such that they can be performed independently in m and n ways respectively. Then m or n occur in $m+n$ ways

(b) Multiplication Principle :

If a first job can be done in m ways and a second job can be done in n ways. Then both the jobs can occur in $m \times n$ ways.

Factorial Notation:

The product of the first n natural positive integers is usually denoted by the symbol n!

$$n! = n(n-1),...,3.2.1$$

Example: 4! = 1.2.3.4 = 24

5! = 1.2.3.4.5 = 120.

Note: 0! = 1.

Deductions:

1. Put r = n in (1), nPn = n!
2. Put r = 0 in (1), $nP_0 = 1$.
3. Put r = 1 in (1) $nP_1 = n$

Permutations in a linear or a circle:

Linear permutation refers to the number of ordered arrangement of objects in a line while circular permutations are an ordered arrangement of objects in a circular manner.

Note:

- The number of ways in which 'n' different things can be arranged in a circle taken all at a time is (n-1)!
- The total number of circular arrangements can be split up into two types:
  (i) Clockwise (ii) Anti-Clockwise
- If the Clockwise and Anti-Clockwise orders are not distinguished, the number of circular permutations is $\frac{(n-1)!}{2}$

- The number of ways in which ‘n’ persons can be seated at a round table is n! or (n-1)! according as the arrangements are made with respect to the table or with respect to each other.

Combinations (or) selections:

Combination means selection of things. The word selection is used, when the order of things has no importance.

Meaning of nCr:

The number of ways of selecting ‘r’ things out of ‘n’ things is denoted by nC_r (or) C (n,r).

Example:

1. Find the value of $10C_3$.

Solution:

$$10C_3 = \frac{10\,(9-1)(9-2)}{1.2.3} = \frac{10.\,9.\,8}{1.\,2.3} = 120$$

Formula $$nCr = \frac{n!}{r!(n-r)!} \qquad (1)$$

Deductions:

1. If r = n in (1) then $nC_n = \frac{n!}{n!(n-n)!} = \frac{n!}{n!} = 1$
2. If r = 1 in (1) then $nC_1 = n$.
3. If r = 0 in (1) then $nC_0 = 1$.

Note

- Number of combinations of ‘n’ things taken ‘r’ at a time is equal to number of things taken n-r at a time.

$$nC_r = nC_{n-r}$$

- $(n+1)C_r = nC_r + nC_{r+1}$

Distinction between Permutation and Combination:

- Permutation means arrangement and combination means selection.
- The order of the things is important in permutation and order of the things has no importance in combination
- Usually the number of permutation exceeds the number of combinations.
- Each combination corresponds too many permutations.

Relation between nP_r and nC_r:

$$nC_r = \frac{nP_r}{r!}$$

LECTURE SCHEDULE- 2

ARITHMETIC, GEOMETRIC AND HARMONIC PROGRESSIONS

Sequence:

A set of numbers arranged in a definite order according to some rule is called a sequence.

Examples:

1, 3, 5, 7……. (Adding 2 to every term)

1, 4, 16, 64… (multiplying by 4 to every term)

Finite sequence

A sequence is called finite if the number of terms is finite. A finite sequence has always a last term.

Example:

2, 5, 8, 11, 14, …, 32

Infinite Sequence

A sequence is called infinite if the number of terms is infinite. An infinite sequence has no last term. In this sequence, every term is followed by a new term.

Examples:

(i) A sequence of multiples of 5

5, 10, 15, 20,…

(ii) A sequence of reciprocals of positive integers

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\ldots$$

Series

Indicated sum of the terms in a sequence is called a series. The result of performing the additions is the sum of the series.

Example:

1+4+7+10+…… is a series in which first term is 1, second term is 4, third term is 7 and so on.

Arithmetic progression (or A.P):

An arithmetic progression is a sequence whose terms increase or decrease by a constant number, called the common difference. The common difference is obtained by subtracting any term by the preceding term.

Notation:

We denote the first term in A.P by 'a' and the common difference by'd'.

Examples:

(i) 1, 3, 5, 7,… form an A.P whose first term is '1' and the common difference is '2'.

General term of an A.P:

a, a + d, a + 2d,…… whose first term is 'a' and the common difference is 'd'.

Note:

- The common difference can also be negative.
- If a = 2; d=3, then the series of quantities in A.P is 2, 5, 8, 11… Thus an A.P is completely known, if its first term and the common difference are known.
- Three numbers in A.P may be taken as a-d, a, a+d where 'a' is not the first term.

The nth term of an A.P:

The nth term of an A.P is $\boxed{t_n = a + (n-1)d}$

If 'n' be the number of terms in the series, the last term 'l' is given by

$$\boxed{l = a + (n-1)\,d}$$

Sum of n terms of an A.P:

Let 'a' be the first term,'d' be the common difference and $l = t_n =$ n^{th} term

To find the sum (S_n) of the n terms of the A.P, we have. $\boxed{S_n = \frac{n}{2}[a + l]}$

Since 'l' is the nth term, we have $l = a + (n-1)\,d$ $\boxed{S_n = \frac{n}{2}[2a + (n-1)d]}$

Properties of an A.P:

If a,b,c,d,…… are in A.P and 'x' is a constant quantity, then the following series.

1. a + x, b + x, c + x, d + x……..
2. a - x, b - x, c - x, d - x,……..
3. ax, bx, cx, dx,……
4. $\frac{a}{x}, \frac{b}{x}, \frac{c}{x}, \frac{d}{x}$, ……. are all in A.P.

Arithmetic mean:

When three quantities are in A.P, the middle one is called the arithmetic mean (A.M) of the other two.

If a, m and b are in A.P, then m is the A.M of a and b, and we have m-a = b-m

(ie) A.M = m = $\frac{a+b}{2}$

Note:

(i) Sum of the first 'n' natural numbers = $\frac{n(n+1)}{2}$

(ii) Sum of the first 'n' natural odd numbers = n^2.

(iii) Sum of the first 'n' natural even numbers = n(n+1)

(iv) Sum of the squares of the first 'n' natural numbers = $\frac{n(n+1)(2n+1)}{6}$

(v) Sum of the cubes of the first 'n' natural numbers = $\left(\frac{n(n+1)}{2}\right)^2$

Geometrical progression (G.P):

A series of quantities are said to be in Geometric progression (or G.P) when each of which after the first is formed by multiplying the successive one by a fixed number, called the common ratio (C.R). The common ratio is obtained by dividing any term by the proceeding one.

Notation:

We denote the first term in G.P by 'a' and the common ratio by 'r'.

General term of a G.P:

a, ar, ar^2,……..

Example:

1, 2, 4, 8, 16,…… are all in G.P whose common ratio is '2'.

Note:

- The common ratio could be a fraction and it might be negative also.
- If a = 2 and r =3, then the numbers in G.P are 2, 6, 18, 54,…..Thus an G.P is completely known ,if its first term and the common ratio are known.
- Three numbers in G.P may always be taken as a/r, a, ar where 'a' is not the first term.

The nth term of a G.P:

Let 'a' be the first term and 'r' be the common ratio of a G.P.

$\therefore$ The nth term of a G.P is $\boxed{t_n = ar^{n-1}}$

If 'l' be the last term of the G.P. containing 'n' terms then $\boxed{l = ar^{n-1}}$

Sum of n terms of a G.P:

Let 'a' be the first term and 'r' be the common ratio of a G.P. Then G.P is a, ar, ar^2, ……

If S_n be the sum of 'n' terms of the G.P, then

$$S_n = \frac{a(r^n - 1)}{r^n - 1}, \; r > 1$$

$$\text{(or)} \quad S_n = \frac{a(1 - r^n)}{1 - r^n}, \; r < 1$$

Sum of an infinite geometric progression series:

$$S_\infty = \frac{a}{1 - r}, \; |r| < 1.$$

Properties of a G.P:

If a,b,c,d,…… are in G.P and 'x' is a constant quantity, then the following series.

1. ax, bx, cx, dx,……
2. $\frac{a}{x}, \frac{b}{x}, \frac{c}{x}, \frac{d}{x}$, ……. are all in G.P.

Geometric Mean:

When three quantities are in G.P, the middle one is called the geometric mean (G.M) of the other two.

If a, m and b are in G.P, then m is the G.M of a and b, and we have $\frac{m}{a}=\frac{b}{m}$

(ie) G.M = m = $\pm\sqrt{ab}$

Harmonic Progression (H.P):

A series of quantities $x_1, x_2, x_3,\ldots\ldots x_n$ is said to be in Harmonic Progression (H.P) when their reciprocals $\frac{1}{x_1},\frac{1}{x_2},\frac{1}{x_3},\ldots\ldots\frac{1}{x_n}$ are in A.P.

Conversely, if the quantities are in A.P, their reciprocals are in H.P.

Thus, if a, b, c are in H.P $\Leftrightarrow$ $\frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in A.P.

Example:

$\frac{1}{4}, \frac{1}{8}, \frac{1}{12},$ ……..are in H.P since 4, 8, 12, …….. are in A.P.

General term of the H.P:

$$\frac{1}{a},\frac{1}{a+d},\frac{1}{a+2d},\ldots\ldots\ldots\ldots$$

The nth term of the H.P:

To find the nth term of the H.P, find the nth term of the corresponding A.P obtained by the reciprocals of the terms of the given H.P. Now the reciprocal of the nth term of an A.P will be the nth term of the H.P (or) $t_n = \frac{1}{a+(n-1)d}$

Note:

There is no general method for finding the sum of the Harmonic Progression.

Harmonic Mean:

When three quantities are in H.P, the middle one is called the Harmonic Mean (H.M) between the other two.

If a, H, b are in H.P, then H is the H.M of a, b and $\frac{1}{a},\frac{1}{H},\frac{1}{b}$ are in A.P.

$$\therefore \quad \frac{1}{H}-\frac{1}{a}=\frac{1}{b}-\frac{1}{H} \Rightarrow \frac{2}{H}=\frac{1}{a}+\frac{1}{b}=\frac{a+b}{ab}$$

$$\Rightarrow \quad \frac{H}{2} = \frac{ab}{a+b} \Rightarrow H = \frac{2ab}{a+b}$$

$$\therefore H.M = H = \frac{2ab}{a+b}$$

Note:

If A, G and H respectively are A.M, G.M and H.M of two positive quantities ‘a’ and ‘b’, then

$$G^2 = A.H \quad \text{and} \quad A \geq G \geq H$$

LECTURE SCHEDULE-3

MATRIX ALGEBRA AND EVALUATION OF DETERMINANTS

Matrix:

A matrix is a rectangular array of numbers arranged in rows and columns enclosed by a pair of brackets [ ].

Example: $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ or $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

Note:

A matrix is usually denoted by a capital letter and its elements by corresponding small letters followed by two suffixes, the first one indicating the row and second one the column in which the element appears.

Order:

A matrix of order m x n (ie. One having m rows and n columns) can be written as

$$A = \begin{pmatrix} a_{11} & a_{12}.....a_{1j}.....a_{1n} \\ a_{21} & a_{22}.....a_{2j}....a_{2n} \\ : & : \quad : \quad : \\ : & : \quad : \quad : \\ a_{i1} & a_{i2}.....a_{ij}.....a_{in} \\ : & : \quad : \quad : \\ : & : \quad : \quad : \\ a_{m1} & a_{m2}...a_{mj}.....a_{mn} \end{pmatrix}$$

where a_{11}, a_{12}, …… stands for real numbers.

Example:

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 1 & 6 \\ 3 & 2 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 0 \\ 5 & 2 & 1 & 3 \end{bmatrix}$$

Here A is of order 3 x 3 and B is of order 3 x 4

Types of matrix:

Row Matrix:

It is a matrix containing only one row and several columns is said to be a row matrix or row vector

Example: $A = \begin{bmatrix} 10.5 & 7.3 & 5.3 & 2.7 \end{bmatrix}$

Column Matrix:

It is a matrix containing only one column. It is also known as column vector.

Example: $A = \begin{bmatrix} 2.3 \\ 1.5 \\ 0.6 \end{bmatrix}$

Square Matrix:

A matrix, in which the number of rows is equal to the number of columns (i.e $m = n$), is called a square matrix.

Example: $A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 1 & 6 \\ 3 & 2 & 2 \end{bmatrix}$

Null (or) Zero Matrix:

A zero matrix or null matrix is a matrix in which all the elements are equal to 0.

Example: $O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Principal Diagonal Matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

The elements a_{11}, a_{22}, a_{33} is called the principal diagonal (or) leading diagonal.

Trace of the matrix

It is defined as the sum of the elements along the leading diagonal.

In this above matrix the trace of the matrix is

$$a_{11} + a_{22} + a_{33}$$

Diagonal Matrix:

It is a square matrix in which all the elements other than in the leading diagonals are zero's.

Example: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$

Scalar Matrix:

In a diagonal matrix, all the diagonal elements are equal; it is called a scalar matrix.

Example: $A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

Unit Matrix (or) Identity Matrix:

If in a diagonal matrix, all the diagonal elements are equal to "1", then the matrix
is called a unit matrix and it is denoted by "I".

Example: $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Upper triangular Matrix:

A square matrix in which all the elements below the leading diagonal are completely zero, then it is said to be an upper triangular matrix.

Example: $A = \begin{bmatrix} 5 & 3 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 4 \end{bmatrix}$

Lower Triangular Matrix:

A square matrix in which all the elements above the leading diagonal are completely zero, then it is said to be a lower triangular matrix.

Example: $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 8 & 2 & 1 \end{bmatrix}$

Sub Matrix:

A matrix obtained by deleting some rows or columns or both of a given matrix is called a sub matrix of a given matrix.

Example: Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 0 \\ 5 & 2 & 1 & 3 \end{bmatrix}$

If we delete the first row and the first column, the sub matrix of A is $\begin{bmatrix} 2 & 1 & 0 \\ 2 & 1 & 3 \end{bmatrix}$

Transpose of a given Matrix:

Let A be a given matrix of order m x n. The matrix obtained by interchanging rows into columns and columns into rows, is called the transpose of the matrix A and it is denoted by A^T of order n x m.

Example: $A = \begin{bmatrix} 2 & 0 & 7 \\ 3 & 2 & 1 \\ 8 & 1 & 0 \end{bmatrix}_{3x3}$; $A^T = \begin{bmatrix} 2 & 3 & 8 \\ 0 & 2 & 1 \\ 7 & 1 & 0 \end{bmatrix}_{3x3}$

Properties of transpose of a matrix

1. $(A^T)^T = A$
2. $(AB)^T = B^T A^T$ is known as the reversal Law of Transpose of product of two matrices.

Symmetric Matrix:

A Square matrix A is said to be symmetric if $A = A^T$.

Example: $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 5 & 7 \\ 4 & 7 & 10 \end{bmatrix}$; $A^T = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 5 & 7 \\ 4 & 7 & 10 \end{bmatrix}$

Skew Symmetric Matrix:

A Square matrix A is said to be skew symmetric if $A = -A^T$.

Example:

$$A = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -3 \\ 2 & 3 & 0 \end{bmatrix} \quad ; \quad A^T = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$$

Here $A = -A^T$

Idempotent Matrix:

A symmetric matrix A is said to be idempotent matrix if $A^2 = A$

Example

$$\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

Singular Matrix:

A square matrix is said to be a singular matrix, if its determinant of A is zero (i.e $|A| = 0$).

Note: If a determinant of A is not equal to zero (i.e $|A| \neq 0$), then the matrix is called non-singular matrix.

Equality of matrices

Two matrices A & B are equal, if and only if,

(i) Both A & B are of the same type

(ii) Every element of 'B' is the same as the corresponding element of 'A'.

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 9 & 10 & -1 & 3 \\ 4 & 2 & 8 & 5 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 10 & 2 \\ 9 & -1 & 3 \\ 4 & 8 & 5 \end{bmatrix}$$

Here though the elements in matrix A are same as the elements in matrix B, the two matrices are not equal.

$$A \neq B$$

Example:

Find the value of a and b given

$$\begin{bmatrix} 4 & 5 \\ a & b \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}$$

Solution:

The given matrices are equal

$\therefore$ a = 3, b = 2

Algebra of Matrices:

Addition of two matrices:

If A = a_{ij} and B = b_{ij} be any two m x n matrices (ie) they have the same number of rows and columns, then their sum A + B is defined by C = c_{ij} where the matrix C is obtained by adding the corresponding elements of A and B.

$$A + B = C = c_{ij} = (a_{ij} + b_{ij})_{m \times n}$$

Example:

Find $\begin{bmatrix} 4 & 5 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$

Solution:

$$\begin{bmatrix} 4 & 5 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 4+2 & 5+3 \\ 5+2 & 6+1 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 7 & 7 \end{bmatrix}$$

Properties of Addition:

1. If A and B are two m x n matrices, then A + B = B + A.
2. If A, B and C be any three m x n matrices, then A + (B + C) = (A + B) + C.
3. For every m x n matrix A, there exists a zero matrix O of the same order, such that A + O = O + A = A.
4. If A and B be any two m x n matrices such that B satisfies the equation A + B = 0, then each element of B is the negative of the corresponding element of A and B is denoted by -A.

Difference of two Matrices:

If A and B be two m x n matrices, A – B = A + (-B), the difference A – B is obtained by subtracting each element of B from the corresponding element of A.

Example:

Find $\begin{bmatrix} 4 & 5 \\ 5 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$

Solution:

$$\begin{bmatrix} 4 & 5 \\ 5 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 4-2 & 5-3 \\ 5-2 & 6-1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix}$$

Multiplication of matrix:

They are of two types: 1. By a scalar ⟶ kB.

2. By a matrix ⟶ A x B

Matrix Multiplication:

Two matrices A & B can be multiplied to form the matrix product AB, if and only if the number of columns of 1st matrix A is equal to the number of rows of 2nd matrix If A is a matrix of order (m x p) and B is a matrix of order (p x n) then the matrix product AB can be formed. AB is a matrix of order (m x n).

In this case the matrices A and B are said to be **conformable** for matrix multiplication.

Example:

Find $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ 2 & -3 \end{bmatrix}$

Solution:

$$\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 2\times 6+3\times 2 & 2\times 4+3\times -3 \\ 4\times 6+5\times 2 & 4\times 4+5\times -3 \end{bmatrix}$$

$$= \begin{bmatrix} 12+6 & 8+(-9) \\ 24+10 & 16+(-15) \end{bmatrix}$$

$$= \begin{bmatrix} 18 & -1 \\ 34 & 1 \end{bmatrix}$$

Note:

The matrix product AB is different from the matrix product BA, for

1. The matrix AB can be formed but not BA

Eg: A is a (2 x 3) matrix

B is a (3 x 5) matrix

AB alone can be formed and it is a (2 x 5) matrix.

2. Even if AB & BA can be formed, they need not be of same order.

Eg: A is a (2 x 3) matrix

B is a (3 x 2) matrix

AB can be formed and is a (2 x 2) matrix

BA can be formed and is a (3x 3) matrix

3. Even if AB & BA are of the same order, they needn't be equal. Because, they need not be identical.

Eg: A is a (3 x 3) matrix

B is a (3 x 3) matrix

AB is a (3 x 3) matrix

BA is a (3x 3) matrix

$AB \neq BA$

Note:

- The multiplication of any matrix with null matrix the resultant matrix is also a null matrix.

- When any matrix (ie.) A is multiplied by unit matrix; the resultant matrix is 'A' itself.

Matrix Multiplication by a Scalar:

Let A be a m x n matrix of order m x n obtained by multiplying each element of the matrix A by the number 'k' is called the scalar multiple of A by 'k', then the obtained matrix is denoted by kA.

Example:

If k = 3 and $A=\begin{bmatrix} 1 & 2 & 5 \\ 2 & -3 & 4 \end{bmatrix}$, then $kA = 3 \begin{bmatrix} 1 & 2 & 5 \\ 2 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 15 \\ 6 & -9 & 12 \end{bmatrix}$

Note:

- If A and B be any two m x n matrices, then $k(A + B) = kA + kB$.
- If A be any matrix of order m x n, then $(-k) A = -kA$.
- $(k_1 + k_2) A = k_1 A + k_2 A$.

Determinants and its Properties:

Definition: Every square matrix has a determinant. A matrix is only an arrangement of numbers where as a determinant has a definite value. If A is a square matrix, its determinant is denoted by $|A|$.

Consider the 2nd order determinant,

$$|A| = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1$$

Consider the 3rd order determinant,

$$|A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Note: Only square matrices have determinants.

Example:

Calculate the determinant of the following matrices.

(*a*) $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{bmatrix}$

Solution:

(a) $\begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix}$ = (3 x 1) – (0 x 2) = 3 – 0 = 3.

(b) $$\begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} = 1\begin{vmatrix} 2 & 4 \\ 3 & -1 \end{vmatrix} - 2\begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} + 3\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix}$$

$$= 1(-2-12) - 2(-3-4) + 3(9-2)$$
$$= -14 + 14 + 21 = 21$$

Properties of Determinants:

1. The value of the determinant remains unaltered when the rows are changed into columns and columns are changed into rows.

Example:

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{bmatrix}$

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} = 1\begin{vmatrix} 2 & 4 \\ 3 & -1 \end{vmatrix} - 2\begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} + 3\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix}$$

$$= 1(-2-12) - 2(-3-4) + 3(9-2)$$

$$= -14 + 14 + 21 = 21$$

Let us interchange the rows and columns of A. Thus we get new matrix A_1.

Then

$$\det(A_1) = \begin{vmatrix} 1 & 3 & 1 \\ 2 & 2 & 3 \\ 3 & 4 & -1 \end{vmatrix} = 1\begin{vmatrix} 2 & 3 \\ 4 & -1 \end{vmatrix} - 3\begin{vmatrix} 2 & 3 \\ 3 & -1 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 3 & 4 \end{vmatrix}$$

$$= 1(-2-12) - 3(-2-9) + 1(8-6)$$

$$= -14 + 33 + 2 = 21$$

Hence $\det(A) = \det(A_1)$.

2. If two rows (or columns) of a determinant are interchanged, then the sign of the determinant is altered (changed).

   (a) For an odd number of changes, the resulting determinant is the negative of the original determinant.

   (b) For an even number of changes, the resulting determinant has the same sign as the original determinant.

Example:

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{bmatrix} \text{ then}$$

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} = 1\begin{vmatrix} 2 & 4 \\ 3 & -1 \end{vmatrix} - 2\begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} + 3\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix}$$

$$= 1(-2-12) - 2(-3-4) + 3(9-2)$$

$$= -14 + 14 + 21 = 21$$

Let A_1 be the matrix obtained from A by interchanging the first and second row. Then

$$\det(A_1) = \begin{vmatrix} 3 & 2 & 4 \\ 1 & 2 & 3 \\ 1 & 3 & -1 \end{vmatrix} = 3\begin{vmatrix} 2 & 3 \\ 3 & -1 \end{vmatrix} - 2\begin{vmatrix} 1 & 3 \\ 1 & -1 \end{vmatrix} + 4\begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix}$$

$$= 3(-2-9) - 2(-1-3) + 4(3-2)$$

$$= -33 + 8 + 4 = -21$$

Hence $\det(A) = -\det(A_1)$.

3. If two rows (or columns) of a determinant be identical, then the value of the determinant is zero.

Example:

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 3 \\ 3 & 3 & 4 \\ 1 & 1 & -1 \end{bmatrix} \text{ then}$$

$$\det(A) = \begin{vmatrix} 1 & 1 & 3 \\ 3 & 3 & 4 \\ 1 & 1 & -1 \end{vmatrix} = 1\begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} - 1\begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} + 3\begin{vmatrix} 3 & 3 \\ 1 & 1 \end{vmatrix}$$

$$= 1(-3-4) - 1(-3-4) + 3(3-3)$$

$$= -7 + 7 + 0 = 0$$

$$\text{Hence } \det(A) = 0$$

4. If each of the element in a row (or column) of a determinant be multiplied by a common number 'k', then the value of determinant is multiplied by 'k'.

Example:

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{bmatrix} \text{ then}$$

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} = 1\begin{vmatrix} 2 & 4 \\ 3 & -1 \end{vmatrix} - 2\begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} + 3\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix}$$

$$= 1(-2-12) - 2(-3-4) + 3(9-2)$$

$$= -14 + 14 + 21 = 21$$

Let A_1 be the matrix obtained by multiplying the elements of the first row by 2 (ie. here k =2) then

$$\det(A_1) = \begin{vmatrix} 2(1) & 2(2) & 2(3) \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} = 2\times 1\begin{vmatrix} 2 & 4 \\ 3 & -1 \end{vmatrix} - 2\times 2\begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} + 2\times 3\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix}$$

$$= 2\left[1(-2-12) - 2(-3-4) + 3(9-2)\right]$$

$$= 2\left[-14 + 14 + 21\right] = 2(21)$$

Hence det (A) = 2 det (A_1).

5. If each of any row (or column) of a determinant is the sum of the two terms, then the determinant can be expressed as the sum of two determinants of the same order.

Example:

$$\text{Let } A = \begin{bmatrix} 1+2 & 2+4 & 3+6 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{bmatrix}$$

Then

$$\det(M_2)=\begin{vmatrix}2 & 4 & 6\\ 3 & 2 & 4\\ 1 & 3 & -1\end{vmatrix}=2\begin{vmatrix}2 & 4\\ 3 & -1\end{vmatrix}-4\begin{vmatrix}3 & 4\\ 1 & -1\end{vmatrix}+6\begin{vmatrix}3 & 2\\ 1 & 3\end{vmatrix}$$

$$=2(-2-12)-4(-3-4)+6(9-2)$$

$$=-28+28+42=42$$

$$\det(A)=\det(M_1)+\det(M_2)$$

Hence $\det(A)=21+42=63$

$$\det(A)=\begin{vmatrix}1+2 & 2+4 & 3+6\\ 3 & 2 & 4\\ 1 & 3 & -1\end{vmatrix}=\begin{vmatrix}1 & 2 & 3\\ 3 & 2 & 4\\ 1 & 3 & -1\end{vmatrix}+\begin{vmatrix}2 & 4 & 6\\ 3 & 2 & 4\\ 1 & 3 & -1\end{vmatrix}$$

$$=\det(M1)+\det(M2)$$

$$\det(M_1)=\begin{vmatrix}1 & 2 & 3\\ 3 & 2 & 4\\ 1 & 3 & -1\end{vmatrix}=1\begin{vmatrix}2 & 4\\ 3 & -1\end{vmatrix}-2\begin{vmatrix}3 & 4\\ 1 & -1\end{vmatrix}+3\begin{vmatrix}3 & 2\\ 1 & 3\end{vmatrix}$$

$$=1(-2-12)-2(-3-4)+3(9-2)$$

$$=-14+14+21=21$$

6. A determinant is unaltered when to each element of any row (column) is added to those of several other rows (columns) multiplied respectively by constant factors.

Example:

Let $A=\begin{bmatrix}1 & 2 & 3\\ 3 & 2 & 4\\ 1 & 3 & -1\end{bmatrix}$ then

$$\det(A)=\begin{vmatrix}1 & 2 & 3\\ 3 & 2 & 4\\ 1 & 3 & -1\end{vmatrix}=1\begin{vmatrix}2 & 4\\ 3 & -1\end{vmatrix}-2\begin{vmatrix}3 & 4\\ 1 & -1\end{vmatrix}+3\begin{vmatrix}3 & 2\\ 1 & 3\end{vmatrix}$$

$$=1(-2-12)-2(-3-4)+3(9-2)$$

$$=-14+14+21=21$$

Let A_1 be a matrix obtained when the elements C_1 of A are added to those of second column and third column multiplied respectively by constants 2 and 3. The

$$\det(A_1) = \begin{vmatrix} 1+2(2)+3(3) & 2 & 3 \\ 3+2(2)+3(4) & 2 & 4 \\ 1+2(3)+3(-1) & 3 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} + \begin{vmatrix} 2(2) & 2 & 3 \\ 2(2) & 2 & 4 \\ 2(3) & 3 & -1 \end{vmatrix} + \begin{vmatrix} 3(3) & 2 & 3 \\ 3(4) & 2 & 4 \\ 3(-1) & 3 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 3 & -1 \end{vmatrix} + 3 \begin{vmatrix} 3 & 2 & & 3 \\ 4 & 2 & 4 & \\ -1 & 3 & & -1 \end{vmatrix}$$

$$= 1(-2-12) - 2(-3-4) + 3(9-2) + 2(0) + 3(0)$$

$$= -14 + 14 + 21 = 21$$

$$\det(A_1) = \begin{vmatrix} 1+2(2)+3(3) & 2 & 3 \\ 3+2(2)+3(4) & 2 & 4 \\ 1+2(3)+3(-1) & 3 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} + \begin{vmatrix} 2(2) & 2 & 3 \\ 2(2) & 2 & 4 \\ 2(3) & 3 & -1 \end{vmatrix} + \begin{vmatrix} 3(3) & 2 & 3 \\ 3(4) & 2 & 4 \\ 3(-1) & 3 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 2 & 3 \\ (2) & 2 & 4 \\ (3) & 3 & -1 \end{vmatrix}$$

LECTURE SCHEDULE-4

INVERSE OF A MATRIX BY ADJOINT METHOD

Inverse Matrix:

Let A be a non-singular square matrix. If there exists a square matrix B, such that BA = AB = I then B is called inverse matrix of A and is denoted by A^{-1}.

Formula:

If A is a non-singular square matrix then the inverse of A, denoted by A^{-1} is equal to

$$A^{-1} = \frac{1}{|A|}\text{adj}\,A,\ |A| \neq 0$$

Working Rule to find the inverse of the matrix

Step 1: Find the determinant of the matrix.

Step 2: If the value of the determinant is non zero proceed to find the inverse of the matrix. ((i.e) only if $|A| \neq 0$ we can find A^{-1})

Step 3: Find the cofactor of each element and form the cofactor matrix.

Step 4: The transpose of the cofactor matrix is the adjoint matrix.

Step 5: The inverse of the matrix $A^{-1} = \frac{adj(A)}{|A|}$

Minor:

The minor of an element in a determinant is the determinant got by suppressing the row and column in which the element appears. The minor of the element in the ith row, j^{th} column is denoted by M_{ij}.

Note: The order of the minor of an element in a determinant A is one less than the order of determinant A.

Cofactor:

The signed minor is called cofactor. The cofactor of the element in the ith row, j^{th} column is denoted by Cij and $C_{ij} = (-1)^{i+j} M_{ij.}$

Note:

The expansion of 3rd order determinant when expanded in terms of minors, the sign of the cofactor of element will be as follows:

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

Adjoint (or) Adjugate of a square Matrix:

The adjoint or adjugate matrix of a square matrix A is the transpose of the cofactor matrix. It is denoted by adj A.

Adjoint of A = (Cofactor of A)^{T}

LECTURE SCHEDULE – 5

SOLUTION OF SIMULTANEOUS EQUATIONS BY CRAMER'S RULE & INVERSE METHOD

Linear Equation:

A first degree equation involving one or two variables is known as a linear equation.

Example:

3x -2y = 9, 4x – 2y = -2 are linear equations in two variables.

Simultaneous Equations:

A pair of linear equations involving two variables taken together is known as Simultaneous equations.

Example:

2x - 3y = -6

3x + 4y = 8

Solution of simultaneous equations using Matrix method:

Let us consider a system of linear equations with three unknowns

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

These set of equations can be written in the matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

(i.e) $AX = B$

(1)

where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$; $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$; $B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

Pre multiplying both sides of (1) by A^{-1}.

$(A^{-1} A)X = A^{-1}B$

We know that $A^{-1} A = A A^{-1} = I$

$\therefore I X = A^{-1}B$ ($\because IX = X$)

Hence the solution is $\boxed{X = A^{-1}B.}$

Solution by Determinant (Cramer's Rule):

Let us consider a system of linear equations with three unknowns

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

Consider the determinant

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$\Delta_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} \qquad \Delta_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$\Delta_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

When $\Delta \neq 0$, the unique solution is given by

$$\boxed{x = \frac{\Delta_x}{\Delta};\ y = \frac{\Delta_y}{\Delta};\ z = \frac{\Delta_z}{\Delta}}$$

Unit II

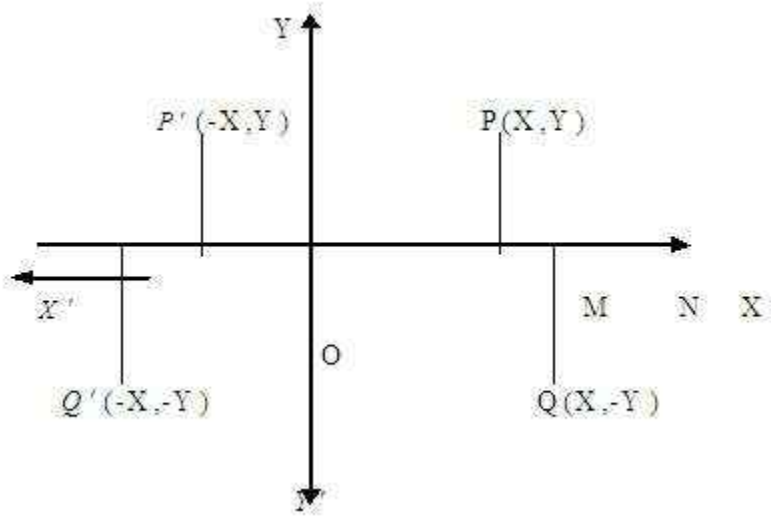
Analytical Geometry

Introduction:

Analytical geometry is the study of geometry using algebraic methods. It is sometimes called Cartesian Geometry or Analytical geometry or Algebraic geometry.

Let X'OX and Y'OY be two perpendicular straight lines intersecting at the point O.

The fixed point O is called origin. The horizontal line X'OX is known as X –axis and the vertical line Y'OY be Y-axis. These two axes divide the entire plane into four parts known as Quadrants.



All the values right of the origin along the X-axis are positive and all the values left of the origin along the X- axis are negative. Similarly all the values above the origin along Y – axis are positive and below the origin are negative.

Let P be any point in the plane .Draw PM perpendicular to X –axis. OM and PM are called X and Y co-ordinates of P respectively and is written as P(X,Y).In particular the origin O has co-ordinates (0,0) and any point on the X-axis has its Y co-ordinate as zero and any point on the Y-axis has its X-co-ordinates as zero

Locus: Locus is the path traced by a moving point under some specified geometrical condition.

Equation of a locus: Any relation in x and y which is satisfied by every point on the locus is called the equation of the locus.

Distance between two points:

The distance between two points $A(x_1,y_1)$ and $B(x_2,y_2)$ is

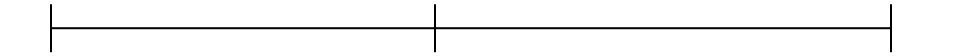
$$\text{dis (AB)} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

In particular the distance between the origin O (0,0) and a point P(x,y) is

$$OP = \sqrt{x^2 + y^2}$$

The internal and External section:

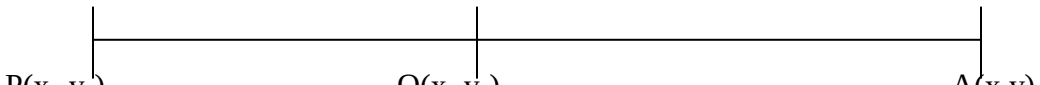
Suppose $P(x_1,y_1)$ and $Q(x_2,y_2)$ are two points



$P(x_1,y_1)$ $A(x,y)$ $Q(x_2,y_2)$

The point A(x,y) that divides distance PQ in the ratio m:n internally is given by

$$A = \left[\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right]$$



$P(x_1,y_1)$ $Q(x_2,y_2)$ $A(x,y)$

The point A(x,y) that divides distance AP and QA in the ratio m : n externally is given by

$$A = \left[\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right]$$

Remark (i) The middle of the line joining (x_1,y_1) and (x_2,y_2) (put m=n= 1) is

$$\left[\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right]$$

(ii) The co- ordinates of point that divides the line joining (x_1,y_1) and (x_2,y_2) in the ratio $\lambda : 1$ are

$$\left[\frac{\lambda x_2 + x_1}{\lambda + 1}, \frac{\lambda y_2 + y_1}{\lambda + 1}\right]$$

(iii) The centroid of a triangle whose vertices are (x_1,y_1,z_1), (x_2,y_2,z_2) and (x_3,y_3,z_3) is given by

$$\left[\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3},\frac{z_1+z_2+z_3}{3}\right]$$

Straight lines

A straight line is the minimum or shortest distance between any two points.

The equation of the coordinate axes

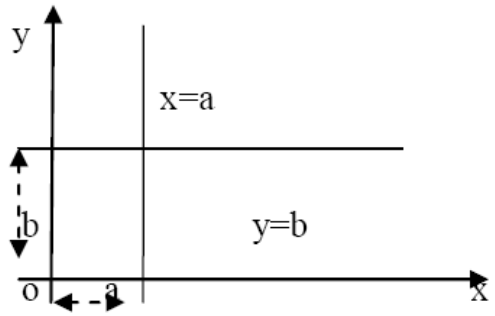
(i) As the y coordinates of every point on the x-axis is zero, the equation of the x-axis is y=0

(ii) As the x coordinates of every point on the y-axis is zero, the equation of the x-axis is x=0

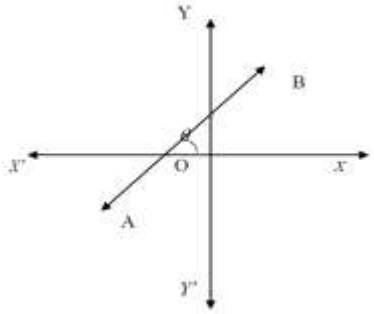
The equation of straight lines parallel to the coordinate axes.

(i) The equation of a line parallel to y axis at a distance „a" from x axis is x=a

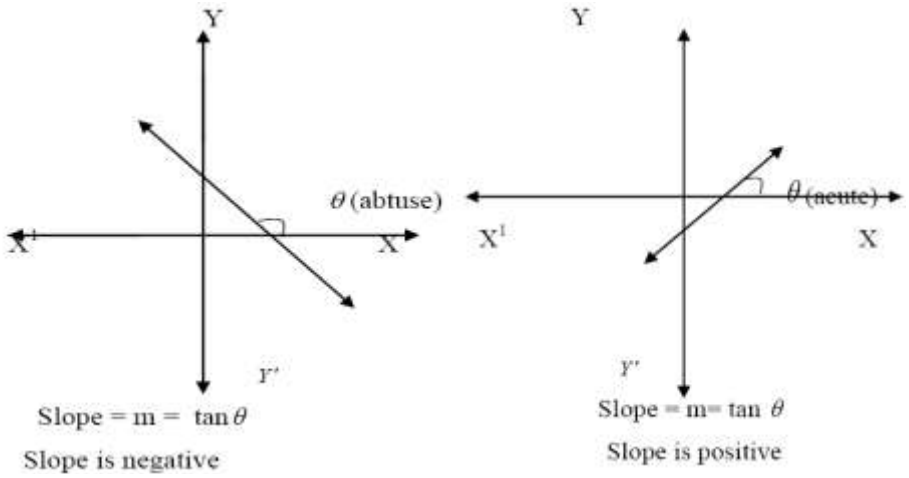
(ii) The equation of a line parallel to x axis at a distance „b" from y axis is y=b



Slope: The slope of the line is the tangent of the angle made by the line with positive direction of X – axis measured in the anticlockwise direction.



let the line AB makes an angle θ with the positive direction of X-axis as in the figure. The angle θ is called the angle of inclination and $\tan\theta$ is slope of the line or gradient of the line. The slope of the line is denoted by m. i.e., slope = m = $\tan\theta$



Note:

(i) The slope of any line parallel to X axis is zero.

(ii) Slope of any line parallel to Y axis is infinity

(iii) The slope of the line joining two points (x1,y1) and (x2,y2) is

$$\text{Slope } m = \tan\theta = \frac{y_2 - y_1}{x_2 - x_1}$$

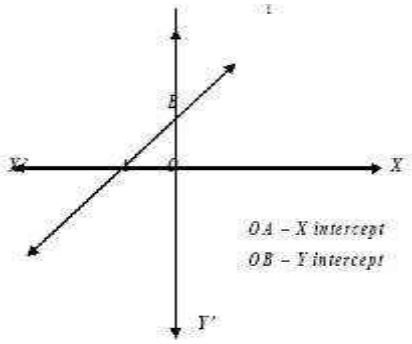
(iv) When two or more lines are parallel then their slopes are equal

(v) When two lines are perpendicular then the product of their slopes is –1

i.e., m1m2 = -1

Equation of Straight lines:

The Equation of any line passing through the origin is of the form Y = mx.



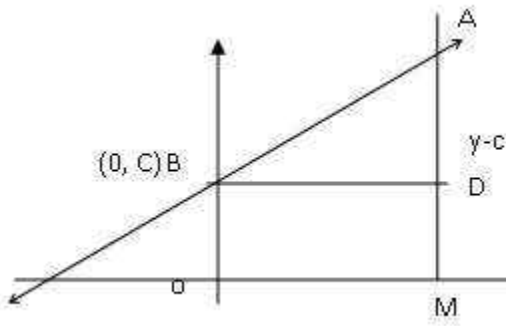
Equation of a straight line: There are several forms of a straight line. They are

- Slope – intercept form
- Slope – one point form

- Two points form
- Intercepts form

1. Slope – intercept form

Let the given line meet y-axis at B (o, c). We call OB as Y – intercept. Let A be any point on the given line. Draw AM perpendicular to OX and BD ⊥ AM. Let this line make an angle θ with X axis. Then the slope,



$$m = \tan\theta = \frac{AD}{BD} = \frac{AM - DM}{OM} = \frac{AM - OB}{OM} = \frac{y - c}{x}$$

ie. $m = \frac{y - c}{x}$

$mx = y - c$

$y = mx + c$

Hence, the equation of a line with slope 'm' and y – intercept 'c' is given by

$y = mx + c$

Note:

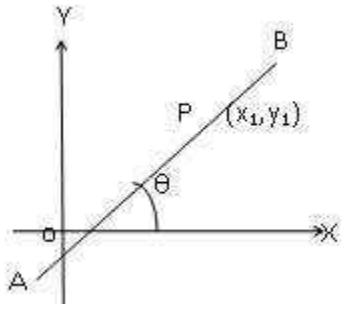
(i) Any line passing through the origin does not cut y – axis.
(c = 0) i.e., y – intercept is zero. Therefore its equation is y = mx

(ii) Any line which is parallel to x – axis has slope equal to zero. Therefore its equation is y = c (Because m = 0)

(iii) Any line perpendicular to x-axis, ie which is parallel to y-axis at a distance of K units from the origin is given by x = k.

2. Slope – one point form

Let the line AB make an angle θ with x- axis as shown in the figure and pass through the point P (x_1, y_1). If (x, y) represents a point other than the point (x_1, y_1), then m = $\frac{y-y_1}{x-x_1}$ where m is the slope of the line or $y - y_1 = m (x - x_1)$.



Hence the equation of a line passing through a point (x_1, y_1) and having slope 'm' is

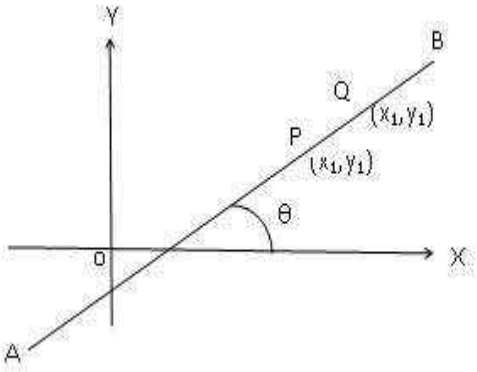
$$y - y_1 = m (x - x_1).$$

3. Two points form

Let P (x_1, y_1) and Q (x_2, y_2) be any two points on the given line AB. We know, the slope, m = $\frac{y_2-y_1}{x_2-x_1}$ We have the slope-point form of a line as

$$y - y_1 = m (x - x_1).$$

Substituting the value of m in the above equation we get,



$$y - y_1 = \frac{y_2-y_1}{x_2-x_1}(x - x_1).$$

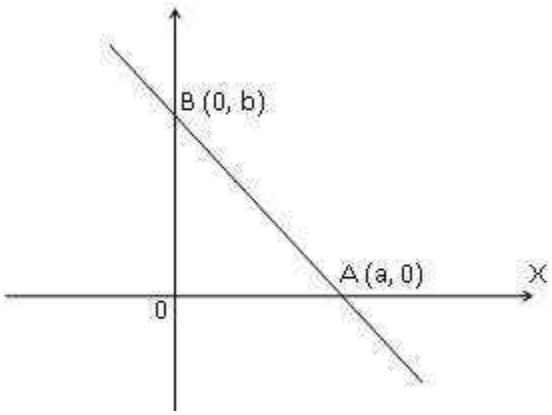
ie $\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$

Hence, the equation of a line passing through two points is given by

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

4. Intercept form

Let AB represent the given line which intersects X – axis at A (a, 0) and Y- axis at B (0, b). We call OA and OB respectively as x and y intercepts of the line.



The two points form of the equation is given by $\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$

Substituting (a, 0) for (x_1,y_1) and (0,b) for (x_2,y_2), we get the equation as

$$\frac{y-0}{b-0} = \frac{x-a}{0-a}$$

ie $$\frac{y}{b} = \frac{x-a}{-a}$$

$$\frac{y}{b} = \frac{x}{-a} - \frac{a}{-a}$$

Thus, $$\frac{y}{b} = \frac{-x}{a} + 1$$

$$\therefore \frac{x}{a} + \frac{y}{b} = 1$$

Hence, the equation of a line having x-intercept ‘a’ and y-intercept ‘b’ is given by

$$\frac{x}{a} + \frac{y}{b} = 1$$

Note: The four equations we have obtained are all first degree equations in x and y. On the other hand it can be shown that the general first degree equation in x and y always represents a straight line. Hence we can take general equation of a straight line as ax + by + c = 0 with at least one of a or b different from c. Further, this gives by = - ax – c

$$\text{i.e.} \quad y = \frac{-ax}{b} - \frac{c}{b}$$

Now, comparing this with the equation y = mx + c, we get

$$\text{slope} = m = \frac{-a}{b} = -\frac{\text{coefficient of } x}{\text{coefficient of } y}$$

Example: 1. Find the equation of a straight line whose (i) Slope is four and y intercept is –3

(ii)Inclination is 30^0 and y intercept is 5

Solution : Slope (m) = 4

Y intercept (c) = -3

Equation of a line is y = mx + c

$$Y = 4x - 3$$

Equation of a line is 4x – y – 3 = 0

(ii) $\theta = 30^0$, y intercept = 5

Slope = $\tan\theta$

$$m = \tan 30^0 = \frac{1}{\sqrt{3}}$$

Equation of a line is y = mx + c

$$Y = \frac{1}{\sqrt{3}}x + 5$$

$$\sqrt{3}\,y = x + 5\sqrt{3}$$

Equation of a line is $x - \sqrt{3}\,y + 5\sqrt{3} = 0$

Example: 2. Calculate the slope and y intercept of the line 2x – 3y + 1= 0

Solution : 2x – 3y + 1 = 0

$$3y = 2x + 1$$

$$y = \frac{2x}{3} + \frac{1}{3}$$

Comparing with y = mx + c, we get

$m = \frac{2}{3}$, $c = \frac{1}{3}$

Slope = $\frac{2}{3}$; y intercept = $\frac{1}{3}$

Example: 3. Find the equation of a straight line passing through (-4,5) and having slope $-\frac{2}{3}$

Solution : Slope = $-\frac{2}{3}$

Point (-4,5)

Equation of the line is $(y-y_1) = m(x-x_1)$

$$y - 5 = -\frac{2}{3}(x+4)$$

$$3y - 15 = -2x - 8$$

$\therefore$ Equation of a line is $2x + 3y - 7 = 0$

Example: 4. Find the equation of the straight line passing through the points (3,6) and (-2,5).

Solution : Equation of the line is $\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$

$$\frac{y-6}{5-6} = \frac{x-3}{-2-3}$$

$$\frac{y-6}{-1} = \frac{x-3}{-5}$$

$$5y - 30 = x - 3$$

$$x - 5y - 3 + 30 = 0$$

Equation of the line is $x - 5y + 27 = 0$

Example.5: Find the intercepts cut off by the line $2x - 3y + 5 = 0$ on the axes.

Solution: x – intercept: put y = 0

$\therefore 2x + 5 = 0$

$x = \frac{-5}{2}$ This is the x – intercept

y – intercept: Put x = 0

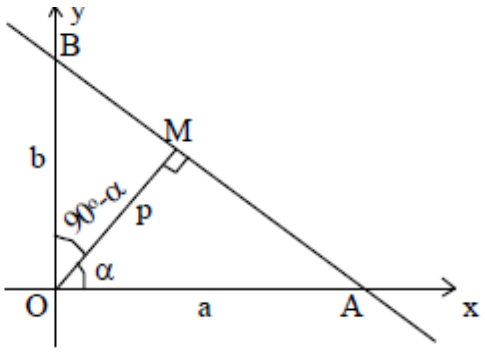
$-3y + 5 = 0$

$\therefore y = \frac{5}{3}$ This is the y – intercept

Normal Form:

When the length of the $\perp$ r from the origin to a straight line is p and the inclination of the $\perp$ r with x -axis is α then the equation of the straight line is

$$x \cos \alpha + y \sin \alpha = p$$



Proof:

Let AB be the line intersects x axis at A and y axis at B.

Let OM $\perp$r AB.

Let OM = p and $\angle XOM = \alpha$.

If the intercepts are a and b then the equation of the straight line is

$$\frac{x}{a} + \frac{y}{b} = 1 \qquad \text{.....(1)}$$

From right angled ΔOAM, $\frac{a}{p} = \sec\alpha \Rightarrow a = p \sec\alpha$

from ΔOBM, $\frac{b}{p} = \text{Sec}(90^\circ-\alpha) \Rightarrow b = p \operatorname{cosec} \alpha$

$\therefore$ (1) $\Rightarrow \frac{x}{p\sec\alpha} + \frac{y}{p\operatorname{cosec}\alpha} = 1$

i.e., $x \cos \boldsymbol{\alpha} + y \sin \boldsymbol{\alpha} = p$ is the equation of a straight line in normal form.

Example

Find the equation of the straight line which has perpendicular distance 5 units from the origin and the inclination of perpendicular with the positive direction of x axis is 120°

Solution:

The equation of the straight line in Normal Form is

$x \cos \alpha + y \sin \alpha = p$

Given $\alpha = 120^\circ$ and $p = 5$

$\therefore$ Equation of the straight line is

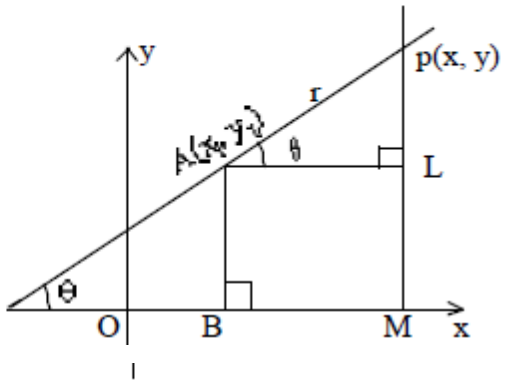
$x \cos 120^\circ + y \sin 120^\circ = 5$

ie $x - y\sqrt{3} + 10 = 0$

Distance form (or) Symmetric form (or) Parametric form General

If the inclination of a straight line passing through a fixed point A with x -axis isθ and any point P on the line is at a distance 'r' from A then its equation is

$$\frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} = r$$



Proof:

Let $A(x_1,y_1)$ be the given point and P(x,y) be any point on the line

AP $= r$,

$\underline{|PAL} = \theta$

Draw PM$\perp$ OX and AL $||$ to x axis.

Then $\cos\theta = \frac{AL}{AP} = \frac{x-x_1}{r}$ and $\sin\theta = \frac{PL}{AP} = \frac{y-y_1}{r}$

$\Rightarrow$ $\frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} = r$ is the required equation.

Any point on this line can be taken as $(x_1 + r\cos\theta, y_1 + r\sin\theta)$ where r is the algebraic distance. Here r is the parameter.

General Form

The equation $ax + by + c = 0$ will always represent a straight line.

Let (x_1, y_1), (x_2, y_2) and (x_3, y_3) be any three points on the locus represented by the equation $ax + by + c = 0$. Then

$$ax_1 + by_1 + c = 0 \quad \ldots (1)$$

$$ax_2 + by_2 + c = 0 \quad \ldots (2)$$

$$ax_3 + by_3 + c = 0 \quad \ldots (3)$$

$(1) \times (y_2 - y_3) + (2) \times (y_3 - y_1) + (3) \times (y_1 - y_2)$ gives

$$a\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right] = 0$$

Since $a \neq 0$, $x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$

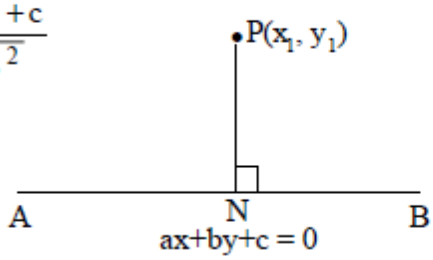
That is (x_1, y_1), (x_2, y_2) and (x_3, y_3) are collinear and hence they lie on a straight line.

Thus the equation $ax + by + c = 0$ represents a straight line.

Observation:

(i) The length of the perpendicular from $P(x_1,y_1)$ to the line $ax+by+c = 0$ is

$$PN = \pm \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}$$



(ii) The length of the perpendicular from the origin to $ax+by+c = 0$ is $\pm \frac{c}{\sqrt{a^2+b^2}}$

Example

Find the length of the perpendicular from (3,2) on the line $3x+2y+1 = 0$

Solution:

Length of the perpendicular from (3,2) to the line $3x+2y+1 = 0$ is

$$\pm \frac{3(3)+2(2)+1}{\sqrt{3^2+2^2}} = \frac{14}{\sqrt{13}}$$

Family of Straight lines: So far we studied about a single straight line. In this section we will discuss about more than one straight line, which lie on a plane.

Intersection of two straight lines The point of intersection of two straight lines is obtained by solving their equations.

Concurrent lines Three or more straight lines are said to be concurrent when they all pass through the same point. That point is known as point of concurrency.

Condition for Concurrency of three lines:

$a_1x+b_1y+c_1 = 0$ .....(i) $a_2x+b_2y+c_2 = 0$ .....(ii)

$a_3x+b_3y+c_3 = 0$ .....(iii) is

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Equation of Straight lines passing through the intersection of two given straight lines

Let $a_1x + b_1y + c_1 = 0$ ... (1)

$a_2x + b_2y + c_2 = 0$ ... (2)

be the equations of the two given straight lines.

Consider the equation $a_1x + b_1y + c_1 + \lambda (a_2x + b_2y + c_2) = 0$ ... (3)

where λ is a constant

Equation (3) is of degree one in x and y and therefore it represents a straight line. Let (x_1, y_1) be the point of intersection of (1) and (2)

$\therefore a_1x_1 + b_1y_1 + c_1 = 0$ and $a_2x_1 + b_2y_1 + c_2 = 0$

$\therefore a_1x_1 + b_1y_1 + c_1 + \lambda (a_2x_1 + b_2y_1 + c_2) = 0$

$\therefore$ Value of (x_1, y_1) satisfies equation (3) also.

Hence $a_1x + b_1y + c_1 + \lambda (a_2x + b_2y + c_2) = 0$ represents a straight line passing through the intersection of the straight lines $a_1x + b_1y + c_1 = 0$ and $a_1x + b_2y + c_2 = 0$

Example: Show that the lines 3x+4y = 13, 2x-7y+1 = 0 and 5x-y=14 are concurrent.

Solution: 3x+4y-13 = 0 2x-7y +1 = 0 and 5x-y -14 = 0

Now,

$$\begin{vmatrix} 3 & 4 & -13 \\ 2 & -7 & 1 \\ 5 & -1 & -14 \end{vmatrix}$$

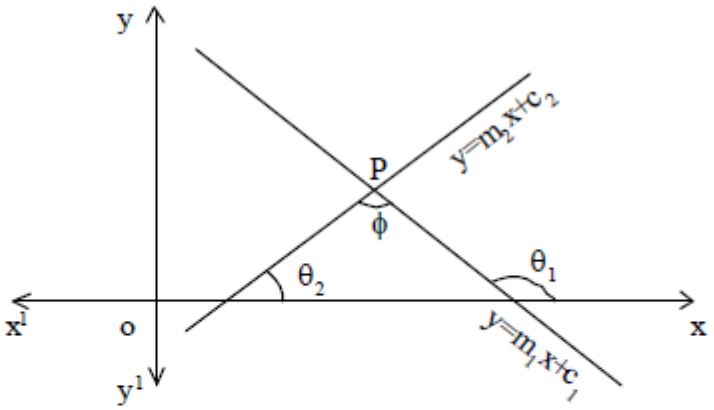
= 3(98+1) - 4 (-28-5)-13 (-2+35)

= 297 + 132-429

= 429-429 = 0

=> the given lines are concurrent

Angle between two straight lines



Let ϕ be the angle between the two straight lines with slopes $m_1 = \tan\theta_1$ and $m_2 = \tan\theta_2$. Then $\tan\phi = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$

$$\therefore \phi = \tan^{-1}\left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

Observation:

(i) If $m_1 = m_2$ the straight lines are parallel
i.e., if the straight lines are parallel then their slopes are equal.

(ii) If $m_1 m_2 = -1$ then the straight lines are $\perp$r. to each other (applicable only when the slopes are finite)
i.e., if the straight lines are perpendicular then the product of their slopes is -1.

Equation of bisectors of the angle between the straight line

Equations of the bisectors of the angles between the straight lines $ax+by+c = 0$ and

$a_1x + b_1y + c_1 = 0$ are $\frac{ax + by + c}{\sqrt{a^2+b^2}} = \pm \frac{a_1x + b_1y + c_1}{\sqrt{a_1^2+b_1^2}}$

Example Find the angle between the straight lines 3x-2y+9=0 and 2x+y-9=0

Slope of the straight line $3x-2y+9=0$ is $m_1=\frac{3}{2}$ $\left[\because y=\frac{3}{2}x+\frac{9}{2}\right]$

Slope of the straight line $2x+y-9=0$ is $m_2=-2$ [$\because y=-2x+9$]

Suppose 'θ' is the angle between the given lines, then

$$\theta = \tan^{-1}\left|\frac{m_1-m_2}{1+m_1 m_2}\right|$$

$$= \tan^{-1}\left|\frac{\frac{3}{2}+2}{1+\frac{3}{2}(-2)}\right| = \tan^{-1}\left|\frac{\frac{7}{2}}{\frac{2-6}{2}}\right|$$

$$= \tan^{-1}\left|-\frac{7}{4}\right| = \tan^{-1}\left(\frac{7}{4}\right)$$

Example:

Find the equation of the bisectors of the angle between 3x+4y+3 = 0 and 4x+3y+1 = 0

Solution:

The equations of the bisectors is $\frac{3x+4y+3}{\sqrt{9+16}} = \pm \frac{4x+3y+1}{\sqrt{16+9}}$

ie., $3x+4y+3 = \pm(4x+3y+1)$

ie., $x-y-2 = 0$ and $7x+7y+4 = 0$

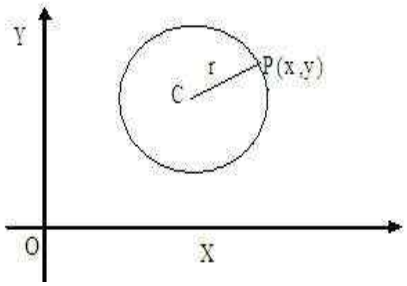
Circle:

A circle is defined as the locus of the point which moves in such a way that its distance from a fixed point is always constant. The fixed point is called Center of the circle and the constant distance is called the radius of the circle.

The equation of the circle when the centre and radius are given :

Let C(h,k) be the centre and r be the radius of the circle. Let P(x,y) be any point on the circle.

CP = r $\Rightarrow$ $CP^2 = r^2$ $\Rightarrow$ $(x-h)^2 + (y-k)^2 = r^2$ is the required equation of the circle.



Note:

If the center of the circle is at the origin i.e., (h,k)=(0,0) then the equation of the circle is $x^2 + y^2 = r^2$

The equation of a circle if the end points of a diameter are given

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the end points of a diameter. Let $P(x, y)$ be any point on the circle.

The angle in a semi circle is a right angle.

$\therefore$ PA is perpendicular to PB

$\therefore$ (Slope of PA) (Slope of PB) $= -1$

$$\left(\frac{y-y_1}{x-x_1}\right)\left(\frac{y-y_2}{x-x_2}\right) = -1$$

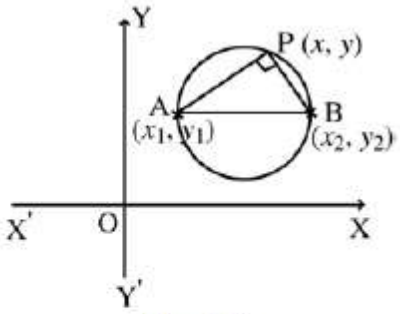


Fig. 5.9.

$$(y-y_1)(y-y_2) = -(x-x_1)(x-x_2)$$

$\therefore (x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$ is the required equation of the circle.

The standard equation of the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Consider the equation $x^2 + y^2 + 2gx + 2fy + c = 0$

This can be written as $x^2 + 2gx + g^2 + y^2 + 2fy + f^2 = g^2 + f^2 - c$

$$(x + g)^2 + (y + f)^2 = \left(\sqrt{g^2 + f^2 - c}\right)^2$$

$$[x-(-g)]^2 + [y-(-f)]^2 = \left(\sqrt{g^2 + f^2 - c}\right)^2$$

This is of the form $(x-h)^2 + (y-k)^2 = r^2$

$\therefore$ The considered equation represents a circle with centre (-g,-f) and radius $\sqrt{g^2 + f^2 - c}$

$\therefore$ The standard equation of the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

C = The Center of the circle whose coordinates are (-g,-f)

R = The radius of the circle = $\sqrt{g^2 + f^2 - c}$

Note:

1. The standard equation of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ possesses the following characteristics.
   i. It is of second degree in x and y.
   ii. The coefficients of x^2 and y^2 are equal to one.

iii. There is no term containing xy.

2. The general equation, $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ always represents a circle.

3. Before finding the center and radius, the general equation $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$----------(1) of the circle must be brought to the standard form. Divide (1) by 'a'

$$x^2 + y^2 + \frac{2g}{a}x + \frac{2f}{a}y + \frac{c}{a} = 0$$ is the standard form.

4. Circles having the same center but different radii are called concentric circles.

Observation :

(i) It is a second degree equation in x and y.

(ii) Coefficient of x^2 = coefficient of y^2

(iii) There is no xy term

(iv) If $g^2+f^2-c > 0$, then circle is a real circle.

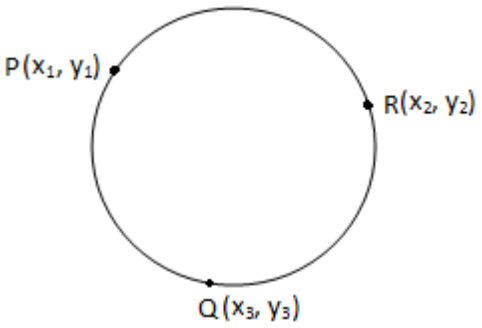
(v) If $g^2+f^2-c = 0$ then circle reduces to a point circle

(vi) If $g^2+f^2-c < 0$ then there is no real circle

(vii) Two or more circles having same centre are called concentric circles.

Equation of the circle passing through three given points

Let P (x1, y1), Q (x2, y2) and R (x3, y3) are the three given points. We have to find the equation of the circle passing through the points P, Q and R.



Let the equation of the general form of the required circle be x2+ y2+ 2gx + 2fy + c = 0 ……. (i) According to the problem, the above equation of the circle passes through the points P(x1, y1), Q (x2, y2) and R (x3, y3).

Therefore, x12 + y12 + 2gx1 + 2fy1+ c = 0 ……. (ii)

x22 + y22 + 2gx2 + 2fy2+ c = 0 ……. (iii)

and x32 + y32 + 2gx3 + 2fy3+ c = 0 ……. (iv)

Form the above there equations (ii), (iii) and (iv) find the value of g, f and c. Then substituting the values of g, f and c in (i) we can find the required equation of the circle.

Example:

Find the equation of the circle with centre at (3, 5) and radius 4 units

Solution:

Equation of the circle whose centre (h, k) and radius r is $(x-h)^2 + (y-k)^2 = r^2$

Given centre (h, k) = (3, 5) and r = 4

$\therefore$ equation of the circle is $(x-3)^2 + (y-5)^2 = 16$

$\Rightarrow$ $x^2+y^2-6x-10y+18 = 0$

Example:

Find the centre and radius of the circle $x^2+y^2-6x+8y-24 = 0$

Solution:

Equation of the circle is $x^2+y^2-6x+8y-24 = 0$

Identifying this with the general form of circle $x^2+y^2+2gx+2fy+c = 0$

we get $2g = -6$; $2f = 8$;

$g = -3$; $f = 4$; $c = -24$

$\therefore$ centre $= (-g, -f) = (3, -4)$

and radius $= \sqrt{g^2 + f^2 - c} = \sqrt{9 + 16 - (-24)} = 7$

Example:

Find the equation of the circle when the coordinates of the end points of the diameter are (3, 2) and (-7, 8)

Solution:

The equation of a circle with end points of diamter as (x_1, y_1) and (x_2, y_2) is

$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$

Here $(x_1, y_1) = (3, 2)$ and

$(x_2, y_2) = (-7, 8)$

$\therefore$ equation of the circle is

$(x-3)(x+7) + (y-2)(y-8) = 0$

$x^2+y^2+4x-10y-5 = 0$

Example: Find the equation the circle passing through the points (0,1), (2,3) and (−2, 5).

Solution:

The general equation of the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

The points (0, 1), (2, 3) and (– 2, 5) lie on the circle

$$\therefore\ 2f + c = -1 \quad \ldots (1)$$

$$4g + 6f + c = -13 \quad \ldots (2)$$

$$-4g + 10f + c = -29 \quad \ldots (3)$$

(1) – (2) $\Rightarrow$ $-4g - 4f = 12$

$$g + f = -3 \quad \ldots (4)$$

(2) – (3) $\Rightarrow$ $8g - 4f = 16$

$$2g - f = 4 \quad \ldots (5)$$

(4) + (5) $\Rightarrow$ $3g = 1 \Rightarrow g = \frac{1}{3}$

(4) $\Rightarrow$ $f = -3 - \frac{1}{3} = -\frac{10}{3}$

(1) $\Rightarrow$ $c = \frac{17}{3}$

$$\therefore x^2 + y^2 + 2\left(\frac{1}{3}\right)x + 2\left(-\frac{10}{3}\right)y + \frac{17}{3} = 0$$

$\therefore 3x^2 + 3y^2 + 2x - 20y + 17 = 0$ is the required equation.

UNIT III

DIFFERENTIAL CALCULUS

Constants, Variables and functions.

A **constant** is a mathematical abbreviation for a number that does not change. A constant can be represented by a Greek letter like α, β (beta) or (more simply) by a letter like a, b or A, B.

A **variable** is something whose magnitude can change i.e something that can take different values. A variable can be represented by x,y,z etc.

A **function** is a mathematical equation that relates two or more variables together. Y=f(x) is a function where x and y are independent and dependent variables.

Limit of a function

Definition

Let f be a function of a real variable x. Let c, l be two fixed numbers. If $f(x)$ approaches the value l as x approaches c, we say l is the limit of the function $f(x)$ as x tends to c. This is written as $\lim_{x \to c} f(x) = l$.

Left Hand and Right Hand Limits

While defining the limit of a function as x tends to c, we consider values of $f(x)$ when x is very close to c. The values of x may be greater or less than c. If we restrict x to values less than c, then we say that x tends to c from below or from the left and write it symbolically as $x \to c - 0$ or simply $x \to c_$. The limit of f with this restriction on x, is called the left hand limit. This is written as

$$Lf(c) = \lim_{x \to c_} f(x), \text{ provided the limit exists.}$$

Similarly if x takes only values greater than c, then x is said to tend to c from above or from right, and is denoted symbolically as $x \to c + 0$ or $x \to c_+$. The limit of f is then called the right hand limit. This is written as $Rf(c) = \lim_{x \to c_+} f(x)$.

It is important to note that for the existence of $\lim_{x \to c} f(x)$ it is necessary that both $Lf(c)$ and $Rf(c)$ exists and $Lf(c) = Rf(c) = \lim_{x \to c} f(x)$. These left and right hand limits are also known as one sided limits.

Fundamental results of Limits Some Important

(1) If $f(x) = k$ for all x, then $\lim_{x \to c} f(x) = k$.

(2) If $f(x) = x$ for all x, then $\lim_{x \to c} f(x) = c$.

(3) If f and g are two functions possessing limits and k is a constant then

(i) $\lim_{x \to c} k f(x) = k \lim_{x \to c} f(x)$

(ii) $\lim_{x \to c} [f(x) + g(x)] = \lim_{x \to c} f(x) + \lim_{x \to c} g(x)$

(iii) $\lim_{x \to c} [f(x) - g(x)] = \lim_{x \to c} f(x) - \lim_{x \to c} g(x)$

(iv) $\lim_{x \to c} [f(x) \cdot g(x)] = \lim_{x \to c} f(x) \cdot \lim_{x \to c} g(x)$

(v) $\lim_{x \to c} \left[\frac{f(x)}{g(x)}\right] = \lim_{x \to c} f(x) \Big/ \lim_{x \to c} g(x), \quad g(x) \neq 0$

(vi) If $f(x) \leq g(x)$ then $\lim_{x \to c} f(x) \leq \lim_{x \to c} g(x)$.

Some Important Limit

For $\left|\frac{\Delta x}{a}\right| < 1$ and for any rational index n,

$$\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1} \qquad (a \neq 0)$$

Solution:

Put $\Delta x = x - a$ so that $\Delta x \to 0$ as $x \to a$ and $\left|\frac{\Delta x}{a}\right| < 1$.

Therefore $$\frac{x^n - a^n}{x - a} = \frac{(a + \Delta x)^n - a^n}{\Delta x} = \frac{a^n \left(1 + \frac{\Delta x}{a}\right)^n - a^n}{\Delta x}$$

$$\left(1+\frac{\Delta x}{a}\right)^n = 1+\binom{n}{1}\left(\frac{\Delta x}{a}\right)+\binom{n}{2}\left(\frac{\Delta x}{a}\right)^2+\binom{n}{3}\left(\frac{\Delta x}{a}\right)^3+\ldots+\binom{n}{r}\left(\frac{\Delta x}{a}\right)^r+\ldots$$

$$\therefore \frac{x^n-a^n}{x-a} = \frac{a^n\left[1+\binom{n}{1}\left(\frac{\Delta x}{a}\right)+\binom{n}{2}\left(\frac{\Delta x}{a}\right)^2+\ldots+\binom{n}{r}\left(\frac{\Delta x}{a}\right)^r+\ldots\right]-a^n}{\Delta x}$$

$$= \frac{\left[\binom{n}{1}a^{n-1}\Delta x+\binom{n}{2}a^{n-2}(\Delta x)^2+\ldots+\binom{n}{r}a^{n-r}(\Delta x)^r+\ldots\right]}{\Delta x}$$

$$= \binom{n}{1}a^{n-1}+\binom{n}{2}a^{n-2}(\Delta x)+\ldots+\binom{n}{r}a^{n-r}(\Delta x)^{r-1}+\ldots$$

$$= \binom{n}{1}a^{n-1}+\text{terms containing } \Delta x \text{ and higher powers of } \Delta x .$$

Since $\Delta x = x-a$, $x \to a$ means $\Delta x \to 0$ and therefore

$$\lim_{x\to a}\frac{x^n-a^n}{x-a} = \lim_{\Delta x\to 0}\binom{n}{1}a^{n-1}+\lim_{\Delta x\to 0}(\text{terms containing } \Delta x \text{ and higher powers of } \Delta x)$$

$$= \binom{n}{1}a^{n-1}+0+0+\ldots = na^{n-1} \qquad \text{since } \binom{n}{1}=n .$$

Limits of trigonometric functions

To evaluate the limits of trigonometric functions, we shall make use of the following limits which are given below:

(i) $\lim_{x\to 0}\frac{\sin x}{x}=1$ (ii) $\lim_{x\to 0}\cos x=1$ (iii) $\lim_{x\to 0}\sin x=0$

Continuity of a Function

Let f be a function defined on an interval I = $[a, b]$. A continuous function on I is a function whose graph $y = f(x)$ can be described by the motion of a particle travelling along it from the point $(a, f(a))$ to the point $(b, f(b))$ without moving off the curve.

Continuity at a point

Definition: A function f is said to be continuous at a point c, $a < c < b$, if

$$\lim_{x\to c} f(x) = f(c)$$

A function f is said to be continuous from the left at c if $\lim_{x\to c^-} f(x) = f(c)$.

Also f is continuous from the right at c if $\lim_{x\to c^+} f(x) = f(c)$. Clearly a function is continuous at c if and only if it is continuous from the left as well as from the right.

Concept of Differentiation

Having defined and studied limits, let us now try and find the exact rates of change at a point. Let us first define and understand what are increments?

Consider a function $y = f(x)$ of a variable x. Suppose x changes from an initial value x_0 to a final value x_1 . Then the increment in x is defined to be the amount of change in x. It is denoted by Δx (read as delta x).That is $\Delta x = x_1 - x_0$.

Thus $x_1 = x_0 + \Delta x$

If x increases then $\Delta x > 0$, since $x_1 > x_0$.

If x decreases then $\Delta x < 0$, since $x_1 < x_0$.

As x changes from x_0 to $x_1 = x_0 + \Delta x$, y changes from $f(x_0)$ to $f(x_0 + \Delta x)$. We put $f(x_0) = y_0$ and $f(x_0 + \Delta x) = y_0 + \Delta y$. The increment in y namely Δy depends on the values of x_0 and Δx. Note that Δy may be either positive, negative or zero (depending on whether y has increased, decreased or remained constant when x changes from x_0 to x_1).

If the increment Δy is divided by Δx, the quotient $\frac{\Delta y}{\Delta x}$ is called the average rate of change of y with respect to x, as x changes from x_0 to $x_0 + \Delta x$. The quotient is given by

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

This fraction is also called a difference quotient.

Definition

The derivative of a given function $y = f(x)$ is defined as the limit of the ratio of the increment Δy of the function to the corresponding increment Δx of the independent variable, when the latter tends to zero.

The symbols y' or $f'(x)$ or $\frac{dy}{dx}$ are used to denote derivative:

$$\frac{dy}{dx} = y' = f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

It is possible for the above limit, not to exist in which case the derivative does not exist. We say that the function $y = f(x)$ is differentiable if it has a derivative.

Derivatives of elementary functions from first principles

I. The derivative of a constant function is zero.

That is, $\frac{d}{dx}(c) = 0$, where c is a constant ... (1)

Proof. Let $f(x) = c$ Then $f(x + \Delta x) = c$

$$\frac{df(x)}{dx} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\therefore \frac{d}{dx}(c) = \lim_{\Delta x \to 0} \frac{c - c}{\Delta x} = 0 .$$

II. The derivative of x^n is nx^{n-1}, where n is a rational number

i.e. $\frac{d}{dx}(x^n) = nx^{n-1}$. ... (2)

Proof: Let $f(x) = x^n$. Then $f(x + \Delta x) = (x + \Delta x)^n$

Now $\frac{df(x)}{dx} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\therefore \frac{d(x^n)}{dx} = \lim_{\Delta x \to 0} \frac{(x + \Delta x)^n - x^n}{\Delta x} = \lim_{\Delta x \to 0} \frac{x^n\left(1 + \frac{\Delta x}{x}\right)^n - x^n}{\Delta x}$$

$$= \lim_{\Delta x \to 0} x^n \left[\frac{\left(1 + \frac{\Delta x}{x}\right)^n - 1}{\Delta x}\right]$$

$$= x^{n-1} \lim_{\Delta x \to 0} \left[\frac{\left(1 + \frac{\Delta x}{x}\right)^n - 1}{\frac{\Delta x}{x}}\right] .$$

Put $y = 1 + \frac{\Delta x}{x}$ As $\Delta x \to 0$, $y \to 1$.

$$\therefore \frac{d(x^n)}{dx} = x^{n-1} \lim_{y \to 1} \left(\frac{y^n - 1}{y - 1}\right)$$

$$= n x^{n-1}$$

$$= nx^{n-1} . \left[\because \lim_{y \to a} \frac{y^n - a^n}{y - a} = na^{n-1}\right]$$

Note. This result is also true for any real number n.

III. The derivative of $\sin x$ is $\cos x$

i.e. if $y = \sin x$ then $\frac{dy}{dx} = \cos x$ ... (3)

Proof:

Let $y = \sin x$. Increase the argument x by Δx, then

$y + \Delta y = \sin(x + \Delta x)$

$$\Delta y = \sin(x + \Delta x) - \sin x = 2 \sin \frac{(x + \Delta x - x)}{2} \cos \frac{(x + \Delta x + x)}{2}$$

$$= 2 \sin \frac{\Delta x}{2} \cdot \cos\left(x + \frac{\Delta x}{2}\right)$$

$$\frac{\Delta y}{\Delta x} = \frac{2 \sin \frac{\Delta x}{2} \cos\left(x + \frac{\Delta x}{2}\right)}{\Delta x} = \frac{\sin \frac{\Delta x}{2}}{\frac{\Delta x}{2}} \cos\left(x + \frac{\Delta x}{2}\right)$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{\sin \frac{\Delta x}{2}}{\frac{\Delta x}{2}} \cdot \lim_{\Delta x \to 0} \cos\left(x + \frac{\Delta x}{2}\right)$$

$$= 1 \cdot \lim_{\Delta x \to 0} \cos\left(x + \frac{\Delta x}{2}\right) .$$

Since $f(x) = \cos x$ is continuous

$$= 1 \cdot \cos x \quad \left| \quad \lim_{\Delta x \to 0} f(x + \Delta x) = \lim_{\Delta x \to 0} \cos(x + \Delta x) \right.$$

$$= \cos x . \quad \left| \quad = \cos x \right.$$

IV. The derivative of $\cos x$ is $-\sin x$

ie. if $y = \cos x$, then $\frac{dy}{dx} = -\sin x$ . ... (4)

Proof: Let $y = \cos x$ Increase the argument x by the increment Δx.

Then $y + \Delta y = \cos(x + \Delta x)$;

$$\Delta y = \cos(x + \Delta x) - \cos x$$

$$= -2 \sin \frac{x + \Delta x - x}{2} \sin \frac{x + \Delta x + x}{2}$$

$$= -2 \sin \frac{\Delta x}{2} \sin\left(x + \frac{\Delta x}{2}\right)$$

$$\frac{\Delta y}{\Delta x} = -\frac{\sin \frac{\Delta x}{2}}{\frac{\Delta x}{2}} \cdot \sin\left(x + \frac{\Delta x}{2}\right) ;$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = -\lim_{\Delta x \to 0} \frac{\sin\frac{\Delta x}{2}}{\frac{\Delta x}{2}} \sin\left(x+\frac{\Delta x}{2}\right)$$

$$= -\lim_{\Delta x \to 0} \frac{\sin\frac{\Delta x}{2}}{\frac{\Delta x}{2}} \cdot \lim_{\Delta x \to 0} \sin\left(x+\frac{\Delta x}{2}\right)$$

Since $\sin x$ is continuous, $\lim_{\Delta x \to 0} \sin\left(x+\frac{\Delta x}{2}\right) = \sin x$ and $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$

$$\therefore \quad \frac{dy}{dx} = -\sin x .$$

V. The derivative of e^x is e^x

Let $y = e^x$. Corresponding to an increment Δx in x, let the increment in y be Δy.

Then $y+\Delta y = e^{x+\Delta x}$

$\therefore \Delta y = e^{x+\Delta x} - e^x$

$$\frac{dy}{dx} = \underset{\Delta x \to 0}{Lt} \frac{\Delta y}{\Delta x}$$

$$= \underset{\Delta x \to 0}{Lt} \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

$$= \underset{\Delta x \to 0}{Lt} \frac{e^x\left(e^{\Delta x} - 1\right)}{\Delta x}$$

$$= e^x \underset{\Delta x \to 0}{Lt} \frac{\left(e^{\Delta x} - 1\right)}{\Delta x}$$

$$= e^x$$

Derivatives of Inverse Trigonometric Functions

I. The derivative of $y = \sin^{-1}x$ is $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

Proof: We have $y = \sin^{-1}x$ and $x = \sin y$

Then $\frac{dx}{dy} = \cos y = \sqrt{1-\sin^2 y} = \sqrt{1-x^2}$

$$\frac{d(\sin^{-1}x)}{dx} = \frac{dy}{dx} = \frac{1}{\left(\frac{dx}{dy}\right)} = \frac{1}{\sqrt{1-x^2}} .$$

II. The derivative of $y = \cos^{-1}x$ is $\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$

Proof: We have $y = \cos^{-1}x$ and $x = \cos y$

$$\therefore \quad \frac{dx}{dy} = -\sin y = -\sqrt{1-\cos^2 y} = -\sqrt{1-x^2}$$

$$\frac{d(\cos^{-1}x)}{dx} = \frac{dy}{dx} = \frac{1}{\left(\frac{dx}{dy}\right)} = \frac{-1}{\sqrt{1-x^2}} .$$

III. The derivative of the function $y = \tan^{-1}x$ is $\frac{dy}{dx} = \frac{1}{1+x^2}$

Proof: We have $y = \tan^{-1}x$ and $x = \tan y$

This implies $\quad x' = \frac{d}{dy}(\tan y) = \sec^2 y = 1 + \tan^2 y = 1 + x^2$

$$y' = \frac{1}{x'} = \frac{1}{1+x^2}$$

IV. The derivative of $y = \cot^{-1}x$ is $y' = -\frac{1}{1+x^2}$.

Proof: We have $y = \cot^{-1}x$ and $x = \cot y$.

$$\frac{dx}{dy} = -\operatorname{cosec}^2 y = -(1+\cot^2 y) = -(1+x^2)$$

$\therefore$ by (20), $\quad \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = -\frac{1}{1+x^2}$.

Aliter : We know that $\quad \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$.

Differentiating with respect to x on both sides,

$$\frac{d(\tan^{-1}x)}{dx} + \frac{d(\cot^{-1}x)}{dx} = \frac{d\left(\frac{\pi}{2}\right)}{dx}$$

$$\frac{1}{1+x^2} + \frac{d(\cot^{-1}x)}{dx} = 0$$

$$\therefore \frac{d(\cot^{-1}x)}{dx} = -\frac{1}{1+x^2} .$$

V. The derivative of $y = \sec^{-1}x$ is $\frac{dy}{dx} = \frac{1}{x\sqrt{x^2-1}}$

Proof: We have $\quad y = \sec^{-1}x$ and $x = \sec y$

$$\frac{dx}{dy} = \sec y \tan y = \sec y \sqrt{\sec^2 y - 1}$$

$$\frac{d(\sec^{-1}x)}{dx} = \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{x\sqrt{x^2-1}}.$$

VI. The derivative of $y = \text{cosec}^{-1}x$ is $\frac{dy}{dx} = -\frac{1}{x\sqrt{x^2-1}}$

Proof: We have $y = \text{cosec}^{-1}x$ and $x = \text{cosec}\, y$

$$\frac{dx}{dy} = \frac{d(\text{cosec}\, y)}{dy} = -\text{cosec}\, y \cot y$$

$$= -\text{cosec}\, y\sqrt{\text{cosec}^2 y - 1} = -x\sqrt{x^2-1}$$

Therefore by (20) $$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = -\frac{1}{x\sqrt{x^2-1}}.$$

Example:

Differentiate $y = \sin^{-1}(x^2 + 2x)$ with respect to x.

Solution: We have $y = \sin^{-1}(x^2 + 2x)$

Take $u = x^2 + 2x$ Then $y = \sin^{-1}(u)$, a function of function.

Therefore by chain rule,

$$y' = \frac{dy}{du}\frac{du}{dx} = \frac{1}{\sqrt{1-u^2}}\frac{d(x^2+2x)}{dx}, \text{ by (21)}$$

$$= \frac{1}{\sqrt{1-(x^2+2x)^2}}(2x+2) = \frac{2(x+1)}{\sqrt{1-x^2(x+2)^2}}.$$

UNIT-II

DIFFERENTIAL CALCULUS-I

LECTURE SCHEDULE – 6

DIFFERENTIATION – DEFINITION – METHODS OF DIFFERENTIATION

Definition (Differentiation)

It is a process of finding the derivative of a function

Ex;

$$D\left(x^2\right) = 2x$$

Differential calculus deals with in finding the rate of change of one quantity with respect to another. Integral calculus deals with in finding a function when its rate of change is known.

Variable and constant

The variable is one which takes different values, whereas a constant takes a fixed value.

Independent and Dependent Variables:

An independent variable is one which can assume any value that is assigned to it. An dependent variable that depends on one or more other variables.

Example

$y = 3x - 2$, the value of y depends on the value chosen for x

'y' –Dependent variable

'x'- Independent variable

Function:

When two or more variables are connected by a relation, each is said to be a function of the other.

Let us consider the growth of a plant. It is a function of time't' (ie) w = f (t) where w = dry matter accumulation.

Here 'w' is the dependent variable and't' is the independent variable.

Even Function:

A function f(x) is said to be even, if f (- x) = f(x) for all values of 'x'.

Example:

$f(x) = x^2$, $f(x) = \cos x$ are even functions.

Odd Function:

A function f(x) is said to be odd, if f (- x) = - f(x) for all values of 'x'.

Example:

$f(x) = x^3$, $f(x) = \sin x$ are odd functions.

Limits:

If f is a function then we say that $\lim_{x \to a} f(x) = l$. If the value of f(x) gets arbitrarily close to '*l*' as x gets closer and closer to '*a*'

Differentiation Formulae

Function(y)	**Derivative** $\left(\frac{dy}{dx}\right)$
x^n	$n\, x^{n-1}$
c	0
$\frac{1}{x^n}$	$\frac{-n}{x^{n+1}}$
e^x	e^x
$\log x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$

cot x	$-\operatorname{cosec}^2 x$
sec x	sec x tan x
cosec x	- cosec x cot x
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\cot^{-1} x$	$-\frac{1}{1+x^2}$
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$
$\operatorname{cosec}^{-1} x$	$-\frac{1}{x\sqrt{x^2-1}}$

Example:

Differentiate following function $g(t) = 2t^6 + 7t^{-6}$

Solution:

Here is the derivative.

$$g'(t) = 2(6)t^5 + 7(-6)t^{-7}$$
$$= 12t^5 - 42t^{-7}$$

Example:

Differentiate following function $y = 8z^3 - \frac{1}{3z^5} + z - 23$

Solution:

$$y = 8z^3 - \frac{1}{3}z^{-5} + z - 23$$

Diff. w.r.to x,

$$y' = 24z^2 + \frac{5}{3}z^{-6} + 1$$

Rules of Differentiation:

1. Differentiation of a sum:

Let y = u + v + w +..... where u,v,w ,..... are functions of x.

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx} + ...$$

2. Differentiation of a difference:

Let y = u – v – w - .....where u,v,w,..... are functions of x.

$$\frac{dy}{dx} = \frac{du}{dx} - \frac{dv}{dx} - \frac{dw}{dx} - .....$$

3. Product Rule of Differentiation:

Let y = uv, where u and v are functions of x.

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Note: If y = uvw, then

$$\frac{dy}{dx} = (uv)\frac{dw}{dx} + (uw)\frac{dv}{dx} + (vw)\frac{du}{dx}$$

Example:

Differentiate f the following functions.

a) $y = \sqrt[3]{x^2}\left(2x - x^2\right)$

Solution:

$$y = \sqrt[3]{x^2}\left(2x - x^2\right)$$

$$y = x^{\frac{2}{3}}\left(2x - x^2\right)$$

Differentiate 'y' with respect to 'x '

$$y' = \frac{2}{3}x^{-\frac{1}{3}}\left(2x - x^2\right) + x^{\frac{2}{3}}\left(2 - 2x\right)$$

Example:

Differentiate the following functions

(b) $f(x) = \left(6x^3 - x\right)\left(10 - 20x\right)$

Solution:

$$f(x) = \left(6x^3 - x\right)\left(10 - 20x\right)$$

Diff f(x) w. r. to 'x'

$$f'(x) = \left(18x^2 - 1\right)\left(10 - 20x\right) + \left(6x^3 - x\right)\left(-20\right)$$
$$= -480x^3 + 180x^2 + 40x - 10$$

4. <u>Quotient Rule:</u>

If $y = \frac{u}{v}$, then

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Example:

Differentiate each of the following functions $P(t) = \frac{\sin(t)}{3 - 2\cos(t)}$

Solution:

In this part we'll need to use the quotient rule.

$$P'(t) = \frac{\cos(t)\left(3 - 2\cos(t)\right) - \sin(t)\left(2\sin(t)\right)}{\left(3 - 2\cos(t)\right)^2}$$
$$= \frac{3\cos(t) - 2\cos^2(t) - 2\sin^2(t)}{\left(3 - 2\cos(t)\right)^2}$$

5. Differentiation of cf(x):

$$\frac{d}{dx}(cf(x)) = c\frac{d}{dx}(f(x) = cf'(x)$$

6. Differentiation of function of a function- Chain Rule:

Let y = f(g(x)). If we have y = f(u) and u=g(x) then the derivative of y is

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Methods of Differentiation:

1. Differentiation of explicit function.
2. Differentiation of Implicit function.
3. Differentiation of Parametric function.
4. Logarithmic differentiation.

1. Differentiation of explicit function:

A function is of the form y = f(x) (or) w = f (t) is called explicit function. To find the differential coefficient (or) derivative for those explicit function it is necessary to differentiate the function with respect to 'x' (or) 't' on both sides

$$\text{(ie)} \quad \frac{dy}{dx} = f'(x) \quad \text{(or)} \quad \frac{dw}{dt} = f'(t)$$

2. Differentiation of Implicit function:

A function is of the form f(x,y) = 0 is called an implicit function. To find the differential coefficient it is necessary to differentiate the function with respect to 'x' on both sides and group the terms containing $\frac{dy}{dx}$ on one side and the other terms on the other side.

Example:

Find $\frac{dy}{dx}$ if $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Solution:

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Differentiating w.r.t 'x'

$$\frac{1}{a^2}\frac{d}{dx}(x^2)+\frac{1}{b^2}\frac{d}{dx}(y^2)=1$$

$$\therefore \frac{1}{a^2}(2x)+\frac{1}{b^2}(2y)\frac{dy}{dx}=0$$

Dividing by 2, $\frac{x}{a^2}+\frac{y}{b^2}\frac{dy}{dx}=0 \Rightarrow \frac{y}{b^2}\frac{dy}{dx}=-\frac{x}{a^2}$

$$\therefore \frac{dy}{dx}=-\frac{b^2x}{a^2y}$$

3. Differentiation of Parametric function:

Let y = f(x) be the given function. If the variables 'x' and 'y' are represented by a new third variable namely 'θ' (or) 't' then x and y are called parametric function and the new variable 'θ' (or) 't' is called parameter.

(ie) $x = f(\theta)$ and $y = g(\theta)$

$$\frac{dx}{d\theta}=f'(\theta) \quad \text{and} \quad \frac{dy}{d\theta}=g'(\theta)$$

$$\frac{dy}{dx}=\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}=\frac{\text{Diff. of y w.r.t parameter}}{\text{Diff. of x w.r.t parameter}}$$

Example:

If x = a (1 + cos θ), y = b (θ + sin θ), find $\frac{dy}{dx}$.

Solution:

x = a (1 + cos θ)

Differentiate w.r.t 'θ', $\frac{dy}{d\theta} = a(-\sin\theta) = -a\sin\theta$

y = b (1 + sin θ); $\frac{dy}{d\theta} = b(\theta + \cos\theta) = b(1 + \cos\theta)$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{b(1+\cos\theta)}{-a\sin\theta} = -\frac{2b\cos^2\frac{\theta}{2}}{2a\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = -\frac{b\cos\frac{\theta}{2}}{a\sin\frac{\theta}{2}} = -\frac{b}{a}\cot\frac{\theta}{2}$$

4. <u>Logarithmic Differentiation</u>:

For some problems, first by taking logarithms and then differentiating, it is easier to find $\frac{dy}{dx}$. Such process is called logarithmic differentiation.

- If the function appears as a product of many simple functions, then by taking the logarithm the product is converted into a sum. It is now easy to differentiate them.

- If the variable x occurs in the exponent then by taking logarithm, it is reduced to a familiar form to differentiate.

Example:

Differentiate: $y = (x + 1)^2 (x + 2)^3 (x + 3)^4$

Solution:

Taking logarithm on both sides

$$\log y = 2\log(x+1) + 3\log(x+2) + 4\log(x+3)$$

Differentiating on both sides,

$$\frac{1}{y}\frac{dy}{dx} = 2\frac{1}{x+1} + 3\frac{1}{x+2} + 4\frac{1}{x+3}$$

$$\frac{dy}{dx} = (x+1)^2 (x+2)^3 (x+3)^4 \left[\frac{2}{x+1} + \frac{3}{x+2} + \frac{4}{x+3}\right]$$

Geometrical and Physical meaning of the derivative

Physical meaning:

We know from the distance-time formula.

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{time taken}}$$

Let the distance's' travelled in time 'a' be given by the distance – time formula s = f (a) then distance travelled in time a + h is f (a + h). Hence in the time interval between 'a' and 'a + h' the distance travelled is f (a + h) – f (a). In the time interval between a and a + h, the speed is given by

$$\frac{\text{Distance}}{\text{Time}} = \frac{\text{Distance travelled upto time } (a+h) \text{ - Distance travelled upto time a}}{\text{Duration of time interval}}$$

$$= \frac{f(a+h) - f(a)}{h}$$

As h $\rightarrow$ 0, the above quantity becomes the speed at the time a.

$$\text{i.e., velocity at time a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

Thus the relation of the derivative to a function is exactly the same way as the relation of the velocity to the distance travelled by a moving particle.

Geometrical Meaning:

The slope of the tangent to the curve at a point is equal to the derivative of the function at that point

$$\tan\theta = m = f'(a)$$

Let y = f (x) be a function Draw the graph of this function. Let P [a, f (a)] be a point on the curve. Take a neighbouring point Q on the curve. Then the coordinates of Q can be taken as Q [a+h, f(a+h)]. Join the chord PQ. Let it meet the x axis at S. Draw PN, QC $\perp$ to x axis. PR $\perp$ QC.

Let PT be the tangent to the curve at P. Let $\angle$ PSN =θ. Then $\angle$QPR. = θ.

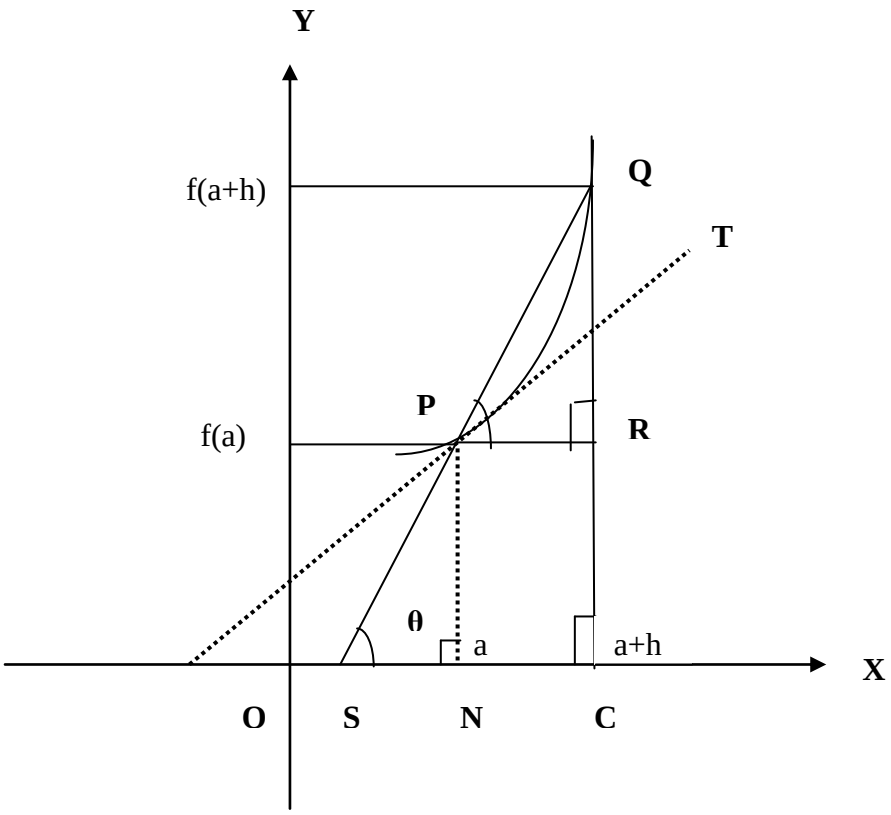
QC=f(a+h), RC=f(a)

Now slope of the chord PQ

$$= \tan\theta = \frac{QR}{PR}$$

$$\therefore \text{Slope of PQ} = \frac{f(a+h)-f(a)}{h}$$

$$\Delta \text{ PQR}, \quad \lim_{\substack{h\to 0 \\ Q\to P}} \frac{QR}{PR} = \tan\theta$$



Chord PQ tends to the tangent at P to the curve $y = f(x)$

LECTURE SCHEDULE – 7

HIGER ORDER DERUVATUVES -– APPLICATIONS OF DIFFERENTIATION

Higher order Differentiation:

Let y = f(x) be the given function. The derivative of 'y' with respect to 'x' is $\frac{dy}{dx}$, called the first derivative.

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) \quad \text{(Second Derivative)}$$

Similarly $\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right)$ called third derivative.

Example:

If $y = (ax+b)^3$; find $\frac{d^3y}{dx}$

Solution:

$y = (ax+b)^3$

Differentiate w.r.t 'x', we get

$$\frac{dy}{dx} = 3(ax+b)^2.a$$

$$\frac{d^2y}{dx^2} = 3.2(ax+b)a^2$$

$$\frac{d^3y}{dx^3} = 3.2.1(a).a^2 = 3!a^3$$

Application of Differentiation:

Growth rate – Relative growth rate, Marginal cost, Marginal revenue

The growth of the plant is usually measured in terms of the dry matter accumulation and is denoted by 'W '. Growth is a function of time t and is denoted by W = f (t) it is called a growth function. Here t is the independent variable and w is the dependent variable.

Similarly absorption is a function of time t and is denoted by A= h (t), it is called a Absorption function. Here t is the independent variable and A is the dependent variable.

Absolute Growth Rate:

Let w = f (t) be the growth function. The derivative of 'W' with respect to't' is called Absolute Growth Rate (AGR) and is denoted by

$$\text{AGR} = \frac{dW}{dt}$$ is the growth rate (or) the absolute growth rate.

(i.e) the rate of change of growth 'W' with respect to time't'.

Absorption rate:

Let A= h (t) be the absorption function. The derivative of 'A' with respect to't' is called Absorption Rate and is denoted by $\frac{dA}{dt}$

Relative Growth rate:

Let W = f (t) be the growth function. The absolute growth is denoted by $\frac{dW}{dt}$. In growth analysis absolute growth rate is not useful concept. The relative growth rate is a measure of the efficiency of plant to produce more. The relative growth rate is defined as

$$\text{RGR} = \frac{\text{Absolute Growth Rate}}{\text{Dry Weight}} = \frac{\text{AGR}}{W} = \frac{dW/dt}{W}$$

$$\boxed{\text{RGR} = \frac{1}{W}.\frac{dW}{dt}}$$

Example:

If $W = e^t + \text{Sint}$. Find the growth rate.

Solution:

$$W = e^t + \text{Sint}$$

$$\frac{dW}{dt} = e^t + \text{cost}$$

Marginal cost and marginal revenue:

Let C = f (Q) be the total cost function and

R = ϕ (Q) be the total revenue function, where Q is the output.

$$\text{Average cost (AC)} = \frac{C}{Q}$$

$$\text{Average Revenue (AR)} = \frac{R}{Q}$$

$$\text{Marginal cost (MC)} = \frac{dC}{dQ}$$

$$\text{Marginal Revenue (MR)} = \frac{dR}{dQ}$$

$$\text{Profit Function } (\pi) = R - C \rightarrow (1)$$

$$\text{Marginal Profit} = \frac{dR}{dQ} - \frac{dC}{dQ}$$

For Profit maximization MR = MC and $\frac{d^2R}{dQ^2} < \frac{d^2C}{dQ^2}$

Example:

The demand function of a firm is P = 500 – 0.2 Q and its cost function is C = 25 Q +10,000. Find the output at which the profits of the firm are maximum. Also find the price it will charge (P = Price; C = Cost ; Q = Output)

Solution:

We know that for maximum profit, MR = MC

Total Cost = (T.C) = 25 Q +10,000

$$\text{Marginal Cost (MC)} = \frac{dc}{dQ} = 25$$

$$\text{Revenue} = R = PQ = (500 - 0.2Q)\,Q = 500Q - 0.2\,Q^2$$

$$\text{Marginal Revenue (MR)} = \frac{dR}{dQ} = 500 - 0.4\,Q$$

For Profit maximization MR = MC $\Rightarrow$ $500 - 0.4\,Q = 25$

(i.e) $0.4\,Q = 475$

$$\therefore\ Q = \frac{475}{0.4} = 1187.5$$

(i.e) Profit maximum output level is Q = 1187.5

The corresponding price $P = 500 - 0.2\,Q = 500 - 0.2\,(1187.50)$ = Rs. 262.50

LECTURE SCHEDULE- 8

PARTIAL DIFFERENTIATION- HOMOGENEOUS FUNCTION AND EULER'S THEOREM

Partial Derivatives:

Let u = f(x,y) be a function of two independent variables x and y. Differentiating u with respect to 'x' keeping 'y' as a constant is known as the partial differential coefficient of 'u'. It is denoted by $\frac{\partial u}{\partial x}$.

$\frac{\partial u}{\partial y}$ means differentiate 'u' with respect to 'y' keeping 'x' constant.

Symbolically, if u = f(x,y) then

$$\frac{\partial u}{\partial x} = \underset{\Delta x \leftarrow 0}{Lim} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$$

$$\frac{\partial u}{\partial y} = \underset{\Delta y \leftarrow 0}{Lim} \frac{f(y, y+\Delta y) - f(x, y)}{\Delta y}$$

Partial Derivatives of Higher Orders:

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial x^2} = u_{xx}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial y \partial x} = u_{yx}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial^2 u}{\partial x \partial y} = u_{xy}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial^2 u}{\partial y^2} = u_{yy}$$

Note:

If u = f(x,y) and its partial derivatives are ***continuous*** then $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$.

Example:

If $u = x^3 + y^3 + 3xy$, find $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$.

Solution:

$$u = x^3 + y^3 + 3xy \qquad (1)$$

Differentiate (1) partially with respect to x, $\frac{\partial u}{\partial x} = 3x^2 + 0 + 3y\,(1) = 3x^2 + 3y.$

Differentiate (1) partially with respect to y, $\frac{\partial u}{\partial y} = 0 + 3y^2 + 3x\,(1) = 3y^2 + 3x.$

Example:

If $f = \tan^{-1}\left(\frac{x}{y}\right)$, verify that $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial y \partial x}$

Solution:

$$f = \tan^{-1}\left(\frac{x}{y}\right) \tag{1}$$

$$\frac{\partial f}{\partial x} = \frac{1}{1+\left(\frac{x}{y}\right)^2}\cdot\left(\frac{1}{y}\right).(1) = \frac{y}{x^2+y^2} \tag{2}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}\left(\frac{y}{x^2+y^2}\right) = \frac{(x^2+y^2)(1)-y(2y)}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2} \tag{3}$$

$$\frac{\partial f}{\partial y} = \frac{1}{1+\left(\frac{x}{y}\right)^2}(x)\left(\frac{-1}{y}\right) = \frac{-x}{x^2+y^2} \tag{4}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{-x}{x^2+y^2}\right) = -\left[\frac{(x^2+y^2)^2(1)-x(2x)}{(x^2+y^2)^2}\right] = \frac{x^2-y^2}{(x^2+y^2)^2} \tag{5}$$

From (3) and (4), $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$

Hence verified

Homogeneous functions:

A function in which each term has the same degree is called a homogeneous function. A function f(x, y) is a homogeneous function of degree n if

$$f(tx, ty) = t^n f(x, y)$$

Example:

$x^2 - 2xy + y^2 = 0$ is a homogenous function of degree 2 in the variables x and y.

Euler's theorem on homogeneous function:

If u=f(x, y, z) is a homogeneous function of degree n in the variables x, y, z ,then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = nu$$

UNIT III

DIFFERENTIAL CALCULUS II:

LECTURE SCHEDULE- 9

INCREASING AND DECREASING FUNCTION- MAXIMA AND MINIMA OF SINGLE VARIABLES- APPLICATIONS IN AGRICULTURE

Increasing and decreasing Functions:

The function y = f(x) is said to be an increasing function of x if as 'x' increases, 'y' also increases. It is said to be a decreasing function of x if as 'x' increases, 'y' decreases and vice versa.

Example:

Consider $y = 3x^2 + 4$

When x = 1, $y = 3(1)^2 + 4 = 7$

When x = 2, $y = 3(2)^2 + 4 = 16$

$\therefore y = 3x^2 + 4$ is an increasing function of x.

Consider $y = 4 – 2x$

When x = 1, $y = 4 – 2 (1) = 2$

When x = 2, $y = 4 – 2 (2) = 0$

$\therefore y = 4 – 2x$ is a decreasing function of x.

Condition for an increasing and a decreasing function:

1. For an increasing function, $\frac{dy}{dx} > 0$
2. For a decreasing function, $\frac{dy}{dx} < 0$.

Example:

For what values of x is the function $y = 8 + 2x – x^2$

(a) increasing (b) decreasing

Solution:

Consider $y = 8 + 2x – x^2$

$$\frac{dy}{dx} = 2- 2x = 2 (1- x)$$

Put $\frac{dy}{dx} = 0 \Rightarrow 2 (1- x) = 0$

(i.e) x = 1

Here when x is greater than one, $\frac{dy}{dx}$ is negative

When x is less than one, $\frac{dy}{dx}$ is positive.

$\therefore$ When $x > 1$, the function $y = 8 + 2x - x^2$ is a decreasing function and

When $x < 1$, the function $y = 8 + 2x - x^2$ is an increasing function.

Maximum point:

If the function y =f (x) increases upto a certain point x = a and then decreases, the point x = a is called a maximum point. The value of y = f (x) at x = a is called a maximum value of the function.

Minimum Point:

If the function y =f (x) decreases upto a certain point x = b and then increases, the point x = b is called a minimum point. The value of y = f (x) at x = b is called a minimum value of the function.

Stationary points:

The maximum and minimum points are also called Stationary points or Turning points or Critical point. The values of the function at the Maximum and Minimum points are called Maximum and Minimum values respectively.

Saddle point: If $f(x, y)$ has neither maximum nor minimum at (x_0, y_0) then such a point is called saddle point

Global maximum and Global minimum:

The maximum of the maximum values is called Global maximum and the minimum of the minimum values is called Global minimum.

Conditions for Maxima and Minima

	First order condition	**Second order condition**
Maximum	$\frac{dy}{dx} = 0$	$\frac{d^2y}{dx^2} < 0$
Minimum	$\frac{dy}{dx} = 0$	$\frac{d^2y}{dx^2} > 0$

Example: Find the Maxima or Minima if any of $x^3 + 2x^2 - 4x - 8$

Solution: Let $y = x^3 + 2x^2 - 4x - 8$----------(1)

$$\frac{dy}{dx} = 3x^2 + 4x - 4, \text{ for maxima and minima}$$

$$\frac{dy}{dx} = 0 \quad \text{(i.e) } 3x^2 + 4x - 4 = 0$$

$$x = \frac{-4 \pm \sqrt{16+48}}{6} = \frac{-4 \pm 8}{6} = 2/3 \text{ or } -2$$

Also $\frac{d^2y}{dx^2} = 6x + 4$

When $x = \frac{d^2y}{dx^2} = 6(-2) + 4 = -8 < 0$

$\therefore$ y is maximum at x = - 2 and the maximum value is obtained by putting x = -2 in (1),

Max $y = (-2)^3 + (2)(-2)^2 - 4(-2) - 8.$

$= -8 + 8 + 8 - 8 = 0$

$\therefore$ Maximum value of the given function is zero.

When x = 2/3, $\frac{d^2y}{dx^2} = 6(2/3) + 4 = 8 > 0.$

$\therefore$ y is minimum at x = 2/3. The minimum value is obtained by putting x = 2/3 in (1).

(i.e) $y = (2/3)^3 + 2(2/3)^2 - 4(2/3) - 1 = \frac{-256}{27}.$

$\therefore$ Minimum value of the given function is $\frac{-256}{27}$.

Physical and Economic optimum

Let y = f (x) be a response function. Here x stands for the input in kgs of fertilizer applied per hectare and y the corresponding output i.e kgs of yield per hectare. We have already seen that the maximum is obtained only when $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$. This optimum is called **physical optimum**, since we are not considering the profit with respect to the investments made.

The necessary condition for the **economic optimum** of the response function is $\frac{dy}{dx} = \frac{P_x}{P_y}$ where P_X = per unit price of input i.e price of fertilizer per kgs

P_Y = per unit price of output i.e price of yield per kgs.

LECTURE SCHEDULE-10

MAXIMA- MINIMA OF SEVERAL VARIABLES WITHOUT CONSTRAINTS

Maximum and Minimum of several variables without Constraints:

Let there be 'n' independent variables $x_1, x_2, \ldots\ldots\ldots x_n$ and the dependent variable be z.

$$\text{Then } z = f(x_1, x_2, \ldots, x_n)$$

The partial derivatives are denoted as follows:

$$\frac{\partial z}{\partial x_1} = f_1, \quad \frac{\partial z}{\partial x_2} = f_2, \ldots\ldots\ldots\ldots\ldots\ldots\ldots \frac{\partial z}{\partial x_n} = f_n$$

$$\frac{\partial^2 z}{\partial x_1^{\,2}} = f_{11}, \ \frac{\partial^2 z}{\partial x_2^{\,2}} = f_{22}, \ldots\ldots\ldots\ldots\ldots\ldots\ldots \frac{\partial^2 z}{\partial x_n^{\,2}} = f_{nn}$$

$$\frac{\partial^2 z}{\partial x_1 \partial x_2} = f_{12}, \ \frac{\partial^2 z}{\partial x_1 \partial x_3} = f_{13}, \ldots\ldots\ldots\ldots\ldots\text{and so on.}$$

In general,

$$\frac{\partial z}{\partial x_i} = f_i, \qquad i = 1,2,3,\ldots\ldots\ldots\ldots n$$

$$\frac{\partial^2 z}{\partial x_i^{\,2}} = f_{ii}, \qquad i = 1,2,3,\ldots\ldots\ldots\ldots n$$

$$\frac{\partial^2 z}{\partial x_i \partial x_j} = f_{ij}, \quad i,j = 1,2,3,\ldots\ldots\ldots.....n$$

The following matrix is called a Hessian matrix:

$$H = \begin{bmatrix} f_{11} & f_{12}.....f_{1n} \\ f_{21} & f_{22}.....f_{2n} \\ ..... \\ ..... \\ f_{n1} & f_{n2}.....f_{nn} \end{bmatrix}$$

Then the minors are

$$H_1 = f_{11}, \qquad H_2 = \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix}, \ H_3 = \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \text{ and so on.}$$

The conditions for Maxima and minima are as follows:

Conditions	Maximum	Minimum
First Order	$f_1 = f_2 = ... = f_n = 0$	$f_1 = f_2 = ... = f_n = 0$
Second Order	$\|H_1\| < 0, \|H_2\| > 0,$ $\|H_3\| < 0, \|H_4\| > 0, ...$	$\|H_1\| > 0, \|H_2\| > 0,$ $\|H_3\| > 0, \|H_4\| > 0, ...$

Working Rule:

Step I : Find the first order derivative and equating to zero and find the values of $x_1, ..., x_n$

Step II: Find the second order derivative to form Hessian matrix

Step III **:** Substitute the values of $x_1, ..., x_n$ in the Hessian Matrix.

Step IV **:** Find the values of the determinant $|H_1|, |H_2|, |H_3|, ...|H_n|$.

If $\{|H_1|, |H_3|, |H_5|\} < 0, \& \{|H_2|, |H_4|, |H_6|\} > 0$ then the function is maximum at $x_1, ..., x_n$

If $\{|H_1|, |H_2|, |H_3|, |H_4|, |H_5|\} > 0$ then the function is minimum at $x_1, ..., x_n$

Step VI **:** Find the Maximum or Minimum values by substituting $x_1, ..., x_n$ in the function z = f ($x_1, ..., x_n$).

UNIT – IV

INTEGRAL CALCULUS

LECTURE SCHEDULE -12

INTEGRATION- METHODS OF INTEGRATION AND DEFINITE INTEGRALS

Integration (Definition)

The process of finding y in $\frac{dy}{dx} = f(x)$ is called integration, we write symbolically

$y = \int f(x)dx$

Integration is the reverse process of differentiation.

Definite Integral (Definition)

If $\int f(x)dx = F(x)$ then $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$ is called the definite integral of $f(x)$ between the limits a and b

Example:

Evaluate $\int_1^2 (x - x^2)dx$

Solution:

$$\int_1^2 (x - x^2)dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_1^2 = \left(\frac{2^2}{2} - \frac{2^3}{3}\right) - \left(\frac{1^2}{2} - \frac{1^3}{3}\right) = 2 - \frac{8}{3} - \frac{1}{2} + \frac{1}{3} = -\frac{5}{6}$$

Indefinite Integral (Definition)

The indefinite integral of $f(x)$ is a function of $F(x)$ whose derivative is equal to $f(x)$. This means that $F'(x) = f(x)$, we write symbolically $\int f(x)dx = F(x) + c$, where c is a constant of integration

Example:

Evaluate $\int 6x^5 - 18x^2 + 7\,dx$

$$\int 6x^5 dx - 18x^2 dx + 7\,dx = \left[6\frac{x^6}{6} - 18\frac{x^3}{3} + 7x\right] + c$$

Integration Formulae

$\int \sin x \, dx$	-cos x + C
$\int \cos x \, dx$	sin x + C
$\int \sec^2 x \, dx$	tan x + C
$\int \operatorname{cosec}^2 x \, dx$	-cot x + C
$\int \sec x \tan x \, dx$	sec x + C
$\int \operatorname{cosec} x \cot x \, dx$	- cosec x + C
$\int \frac{dx}{\sqrt{1-x^2}}$	$\sin^{-1} x$ + C (or) $-\cos^{-1} x$ + C
$\int \frac{dx}{1+x^2}$	$\tan^{-1} x$ + C (or) $-\cot^{-1} x$ + C
$\int \frac{dx}{x\sqrt{x^2-1}}$	$\sec^{-1} x$ + C (or) $-\operatorname{cosec}^{-1} x$ + C
$\int \tan x \, dx$	$\log\lvert\sec x\rvert + C$
$\int \cot x \, dx$	$\log\lvert\sin x\rvert + C$
$\int \sec x \, dx$	$\log\lvert\sec x + \tan x\rvert + C$
$\int \operatorname{cosec} x \, dx$	$-\log\lvert\operatorname{cosec} x + \cot x\rvert + C$
$\int \frac{dx}{x^2+a^2}$	$\frac{1}{a}\tan^{-1}(\frac{x}{a}) + C$
$\int \frac{dx}{x^2-a^2}$	$\frac{1}{2a}\log\left\lvert\frac{x-a}{x+a}\right\rvert + C$
$\int \frac{dx}{a^2-x^2}$	$\frac{1}{2a}\log\left\lvert\frac{a+x}{a+x}\right\rvert + C$
$\int \frac{dx}{\sqrt{a^2-x^2}}$	$\sin^{-1}(\frac{x}{a}) + C$
$\int \frac{dx}{\sqrt{x^2+a^2}}$	$\log\left\lvert x+\sqrt{x^2+a^2}\right\rvert + C$
$\int \sqrt{a^2-x^2} \, dx$	$\frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2}\sin^{-1}(\frac{x}{a}) + C$
$\int \sqrt{x^2+a^2} \, dx$	$\frac{x\sqrt{x^2+a^2}}{2} + \frac{a^2}{2}\log\left\lvert x+\sqrt{x^2+a^2}\right\rvert + C$
$\int \sqrt{x^2-a^2} \, dx$	$\frac{x\sqrt{x^2-a^2}}{2} - \frac{a^2}{2}\log\left\lvert x+\sqrt{x^2-a^2}\right\rvert + C$
$\int dx$	x + C

Basic rules of Integration:

1. $\int cf(x)dx = c\int f(x)dx$ **,** where 'c' is a constant.

2. $\int (f(x) \pm g(x))dx = \int f(x)dx \pm \int g(x)dx$

Example:

$$\int (x^3 + 2x^2 + x + 2)dx = \int x^3dx + 2\int x^2dx + \int xdx + 2\int dx$$

$$= \frac{x^4}{4} + 2\frac{x^3}{3} + \frac{x^2}{2} + 2x + C$$

Methods of Integration:

Integration by Substitution

Integration by Parts

1. Integration by Substitution:

Some of the integrals may not be integrated by applying standard forms, then they can be reduced to one of the standard forms by changing the independent variable. This method of integration is known as integration by substitution.

I. Integrals of the form: $\int f(ax + b)dx$ (Substitution Method)

Method of solving: Put t = ax+b

$$\Rightarrow \frac{dt}{dx} = a$$

$$\Rightarrow dx = \frac{1}{a}dt$$

$$\therefore \int f(ax + b)dx = \int f(t).\frac{1}{a}dt = \frac{1}{a}\int f(t)dt$$

Example:

Evaluate: $\int \text{Sin}(ax + b)\,dx$

Solution:

Put $ax + b = t$

$a\,dx = dt$

$\Rightarrow dx = \frac{1}{a}\,dt$

$$\int \text{Sin}(ax + b)\,dx = \frac{1}{a}\int \text{Sin}\, t\,dt = \frac{1}{a}(-\cos t) + C = -\frac{1}{a}\cos(ax + b) + C$$

II. Integrals of the form: $\frac{\int f'(x)}{\int f(x)}dx$

Method of solving: Put $t = f(x)$

$\Rightarrow dt = f'(x)dx$

$$\therefore \quad \frac{\int f'(x)}{\int f(x)}dx = \int \frac{dt}{t} = \log t + C = \log f(x) + C$$

Example:

Evaluate: $\int \frac{2x}{1 + x^2}dx$

Solution:

Put $1 + x^2 = t$

$2x\,dx = dt$

$dx = \frac{1}{2x}dt$

$$\int \frac{2x}{1 + x^2}dx = \int \frac{1}{t}dt = \log t + C = \log f(x) + C$$

III. Integrals of the form: $\int \frac{f'(x)}{\sqrt{f(x)}}dx$

Method of solving: Put $t = f(x)$

$\Rightarrow dt = f'(x)dx$

$$\therefore \int \frac{f'(x)}{\sqrt{f(x)}}dx = \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} + C = 2\sqrt{f(x)} + C$$

Example:

Evaluate: $\int \frac{\text{Sec}^2 x}{\sqrt{\tan x}} dx$

Solution:

Put $t = \tan x$

$dt = \sec^2 x \, dx$

$$\int \frac{\text{Sec}^2 x}{\sqrt{\tan x}} dx = \int \frac{1}{\sqrt{t}} dt = \int t^{-\frac{1}{2}} dt = \frac{t^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = 2\sqrt{t} + C = 2\sqrt{\tan x} + C$$

IV. Integrals of the form: $\int (f(x))^n f'(x)\, dx$

Method of solving: Put $t = f(x)$

$\Rightarrow dt = f'(x)\, dx$

$$\therefore \int (f(x))^n f'(x)\, dx = \int t^n \, dt = \frac{t^{n+1}}{n+1} + C = \frac{(f(x))^{n+1}}{n+1} + C$$

Example:

Evaluate: $\int (3x^2 + x - 1)^7 (6x + 1)\, dx$

Solution:

Put $t = (3x^2 + x - 1)^7$

$dt = (6x + 1)\, dx$

$$\int (3x^2 + x - 1)^7 (6x + 1)\, dx = \int t^7 dt = \frac{t^8}{8} + C = \frac{1}{8}(3x^2 + x - 1)^8 + C$$

V. Integrals of the form: $\int \frac{dx}{ax^2 + bx + c}$

In this type of integrals if the denominator of integrand is factorisable, it can be split into partial fractions and then can be integrated. Otherwise by writing the denominator of the integrand, as the sum or difference of squares, it can be integrated.

Example:

Integrate: $\int \frac{dx}{x^2 + 3x + 2}$

Solution:

$x^2 + 3x + 2 = (x + 2)(x + 1)$

Apply partial fractions

Let $\frac{1}{(x+1)(x+2)} = \frac{A}{(x+1)} + \frac{B}{(x+2)}$

$1 = A(x+2) + B(x+1)$

Put x = -1, $1 = A(-1+2) + B(-1+1)$

$\boxed{A = 1}$

Put x = -2, $1 = A(-2+2) + B(-2+1)$

$\boxed{B = -1}$

$\therefore \frac{1}{(x+1)(x+2)} = \frac{1}{(x+1)} - \frac{1}{(x+2)}$

$$\int \frac{dx}{x^2+3x+2} = \int \frac{dx}{x+1} - \int \frac{dx}{x+2} = \log(x+1) - \log(x+2) + C = \log\left(\frac{x+1}{x+2}\right) + C$$

VI. Integrals of the form: $\int \frac{dx}{\sqrt{ax^2+bx+c}}$

This type of integrals can be evaluated by writing $ax^2 + bx + c$ as the sum or difference of squares.

Example:

Integrate: $\int \frac{dx}{\sqrt{x^2-4x+2}}$

Solution:

$x^2 - 4x + 2 = x^2 - 4x + 2^2 - 2^2 + 2 = (x-2)^2 - 2$

$$\therefore \int \frac{dx}{\sqrt{x^2-4x+2}} = \int \frac{dx}{\sqrt{(x-2)^2-(\sqrt{2})^2}} = \log\left((x-2) + \sqrt{(x-2)^2-(\sqrt{2})^2}\right) + C$$

$$= \log\left((x-2) + \sqrt{x^2-4x+2}\right) + C$$

LECTURE SCHEDULE-13

INTEGRATION BY PARTS -APPLICATION OF INTEGRATION IN AREA AND VOLUME.

Integration by Parts:

The derivative of the product of two functions is

$$d(uv) = udv + vdu$$

Integrating on both sides, we get

$$\int d(uv) = \int udv + \int vdu$$

$$uv = \int udv + \int vdu$$

$$\therefore \int udv = uv - \int vdu$$

is called the method of integration by parts.

Generalized Formula (Integration by parts)

Bernoulli's Formula

$$\int udv = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \ldots$$

Where u, u', u'', u''' differentiation u successively with respect to x

v, v_1, v_2, v_3 Integration v successively with respect to x

The following table may be helpful in proper selection of u and v:

S.No	Integral	u	dv
1.	$\int x^n \log x \, dx$	logx	$x^n \, dx$
2.	$\int x^n \sin^{-1}x \, dx$	$\sin^{-1}x$	$x^n \, dx$
3.	$\int x^n e^x \, dx$	x^n	$e^x \, dx$
4.	$\int x^n \sin x \, dx$ (or) $\int x^n \cos x \, dx$	x^n	Sinx dx (or) cosx dx
5.	$\int e^x \sin x \, dx$ (or) $\int e^x \cos x \, dx$	e^x (or) sinx (or) cosx	Remaining

Example:

Evaluate: $\int x\, e^x dx$

Solution:

Put $u = x$ and $dv = e^x dx$

$\therefore du = dx$ and $v = e^x$

$$\boxed{\int udv = uv - \int vdu}$$

$$\int x\, e^x dx = x\, e^x - \int e^x dx = x\, e^x - e^x + C = e^x (x - 1) + C.$$

Area:

1. The area enclosed between the curve y = f(x), the x-axis and the ordinates x = a and x = b is $\int_a^b y\, dx$ or $\int_a^b f(x)\, dx$.
2. The area enclosed between the curve x = f(y), the y-axis and the ordinates y = c and y = d is $\int_c^d x\, dy$ or $\int_c^d f(y) dy$.

Volume:

1. The volume of the solid obtained by rotating the area enclosed by y = f(x), the x-axis and the ordinates x = a and x = b is $\pi\int_a^b y^2\, dx$.
2. The volume of the solid obtained by rotating the area enclosed by x = f(y), the y-axis and the ordinates y = c and y = d is $\pi\int_c^d x^2\, dy$.

UNIT – V

MATHEMATICAL MODELS

LECTURE SCHEDULE- 14

AGRICULTURAL SYSTEMS MATHEMATICAL MODELS – CLASSIFICATION OF MATHEMATICAL MODELS

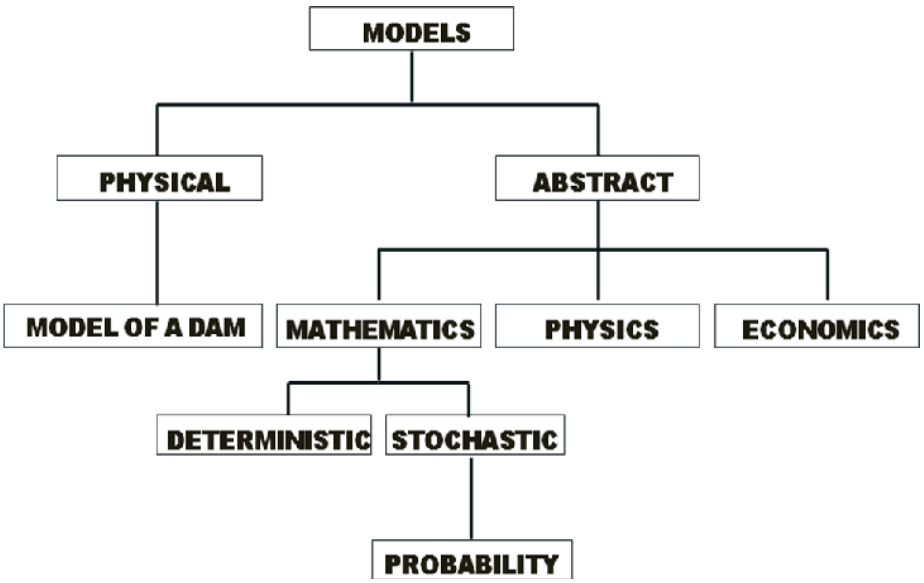
Model

Model is defined as a physical representation of any natural phenomena.

Example: The show case Model exhibited in textile show rooms.

Classification of Models

Models can be broadly classified into physical and abstract. Scale model of a dam, an aircraft, road map of a city etc are all physical models for they physically represent the original. On the contrary, models used in mathematics, economics, biology etc are all abstract in nature. They are conceptual models. Whether physical or abstract, models do not include all the important features of the original. For example, a model of a dam will not exactly tell us what will be the area irrigated in the head and tail reach regions. It will not tell us at what time points the dam will be dry. Similarly a mathematical model that predicts the yield of a crop given the quantity of applied fertilizer may not tell us what will be the most optimal time of application of fertilizer. To answer such questions the modeler has to refine the model again and again.



Mathematical Models

A mathematical model is a description of a system using mathematical concepts and language. This is a representation of phenomena by means of mathematical equations. If the phenomena is the growth, the corresponding Model is said to be Growth Model. A model may help explain a system, study the effects of different components, and make predictions about behavior.

Examples: 1) $W = a + bt$ (Linear model)

2) $W = a + bt + ct^2$ (Quadratic model)

3) $W = \alpha\, \beta^t$ (Exponential model)

are examples of growth models.

Features of Mathematical Models

The mathematical modeling of a real system involves the following distinct phases:

1. Identifying the problem
2. Identifying variables
3. Formulating the model by writing the systems of equations connecting the variables
4. Analyzing the model mathematically
5. Interpreting the model
6. Testing and refining the model with observed data and
7. Concluding and predicting the real world system.

Classification of Mathematical Models

Mathematical models can be broadly divided into two categories, deterministic and stochastic.

In deterministic models random fluctuations are ignored and events associated with the system are assumed to occur in a deterministic way.

Ex.: Predicting the amount of money in a bank account

Stochastic models take into account the random behavior of the system and so events associated with the system are assumed to occur with probability.

Ex. : Wind load, weather conditions

Uses of Mathematical Models

1. Mathematical models are intellectual tools. They help us to define and understand the system under study.
2. Mathematical language differs from natural language in that it is more concise and less ambiguous. It allows us to reason in more complex situations, with less effort, and with less risk of confusion.

3. Mathematical models help us to organize our thoughts and help us to concentrate on the most relevant factors responsible for the system behavior.

4. Mathematical models help us make predictions or simulations. It is the only analytical tool available to make best predictions.

5. Mathematical models provide information on our knowledge gaps and give direction for further research work.

6. Mathematical models can help to understand and explore the meaning of equations or functional relationships.

Types of Growth Models and their equations:

Models

Sl. No.	Name of the Models	The equation of the Models
1.	Linear Model	$W = a \pm bt$
2.	Quadratic Model	$Y = a + bx - cx^2$
3.	Exponential Model	$W = A e^{rt}$
4.	Logistic Model	$W = a / (1 + be^{-ct})$
5.	Mitcherlich's Model	$W = A (1-e^{-bt})$
6.	Gompertz's Model	$W = A\ e^{-e^{\beta - kt}}$
7.	Richard's Model	$W = A (1 \pm e^{\beta - kt})^{-1/n}$

LECTURE SCHEDULE- 15

LINEAR, QUADRATIC, EXPONENTIAL AND LOGISTIC MODELS-THEIR APPLICATIONS IN AGRICULTURE

Linear Model

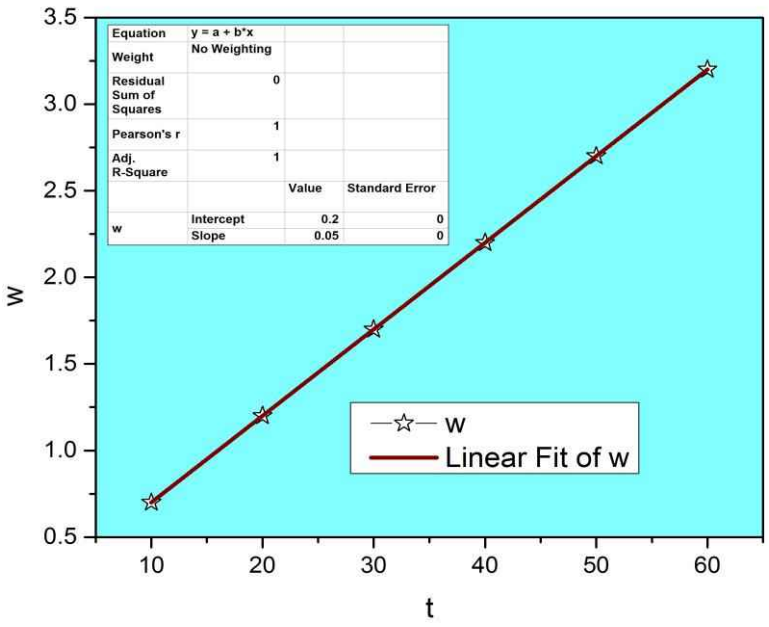
The general form of a linear model is y=a+bx. Here both the variables x and y are of degree 1. In a linear growth model, the dependent variable is always the total dry weight which is noted by w and the independent variable is the time denoted by t. Hence the linear growth model is given by **W = a + bt,**

$$\text{where } a = \overline{W} - b\bar{t}, \overline{W} = \frac{\Sigma W}{n}, \bar{t} = \frac{\Sigma t}{n}, b = \frac{\Sigma tW - \frac{\Sigma t \Sigma W}{n}}{\Sigma t^2 - \frac{(\Sigma t)^2}{n}}$$

Example:

The dry matter accumulation observed on various days for Black gram is given below. Calculate the seed dry weight and growth rate.

t measured in days	10	20	30	40	50	60
W total dry weight in gm/plt	0.7	1.2	1.7	2.2	2.7	3.2



Fitting up of linear models

For fitting up of different models, first we have to investigate whether there is any relationship between the variables, say x and y.

For this purpose we use **Scatter Diagram**,

Let $(x_1, y_1), (x_2, y_2),\ldots\ldots (x_n, y_n)$ be n observations. If the values of the variables x and y are plotted along the x-axis and y-axis respectively in the xy-plane of a graph sheet, the resultant diagram of dots is known as scatter diagram.

The functional relationships may be given by algebraic expressions like

$$\mathbf{Y = a + bx}$$

Simply the line is represented by the equation y = a + bx. For each observed value of the variable x, we can find out the value of y the computed values of y are knows at the expected values of y. If the observed value of y is denoted by y and the expected value by $\hat{y}$, the difference between the y - $\hat{y}$, is knows as error (or) residual the residual is denoted by e. The residuals may be positive (or) negative (or) zero. While fitting a curve, our aim is to fit a best line. The difference between the observed value and the expected value is called a residual. The Principles of least squares states that the constants occurring in the curve of best fit should be chosen such that the sum of the squares of the residuals must be a minimum.

According to the principle of least squares, one should select a and b such that Σei^2 will be as small as possible, that is, we minimize

$$\Sigma ei^2 = S = \Sigma (y - \hat{y})^2$$

$$S = \Sigma (y - (a+bx))^2$$

$$S = \Sigma (y - a - bx)^2$$

To minimize this function first we take the partial derivatives of S with respect to a and b. Then partial derivatives are equated to zero separately. These will result in the equations known as normal equations.

$$S = \sum (y-a-bx)^2$$

$$\frac{\partial S}{\partial a} = 2\sum (y-a-bx)(-1)$$

$$\frac{\partial S}{\partial a} = -2\sum (y-a-bx) = 0$$

$$\sum (y-a-bx)=0$$

$$\sum y - \sum a - b\sum x = 0$$

$$\sum y = na + b\sum x \quad (\because \sum a = na)$$

$$\sum (y-a-bx)^2$$

$$S = \frac{\partial S}{\partial b} = 2\sum (y-a-bx)(-x)$$

$$\frac{\partial S}{\partial b} = -2\sum (y-a-bx) = 0$$

The above equations 1 and 2 are known as normal equations.

From normal equations 1

$$\sum y = na + b\sum x$$

$$\text{divide by } n$$

$$\frac{\sum y}{n} = \frac{na}{n} + \frac{b\sum x}{n}$$

$$\bar{y} = a + b\bar{x}$$

$$a = \bar{y} - b\,\bar{x}$$

From normal equations (2)

$$\sum xy = a\sum x + b\sum x^2$$

$$\sum xy = \left(\frac{\sum y}{n} - \frac{b\sum x}{n}\right)\sum x + b\sum x^2$$

$$\sum xy = \frac{\sum x.\sum y}{n} - \frac{b(\sum x)^2}{n} + b\sum x^2$$

$$\sum xy - \frac{\sum x.\sum y}{n} = \frac{-b(\sum x)^2 + nb\sum x^2}{n}$$

$$\sum xy - \frac{\sum x.\sum y}{n} = b\left[\frac{n\sum x^2 - (\sum x)^2}{n}\right]$$

$$b = \frac{\sum xy - \frac{\sum x.\sum y}{n}}{\frac{n\sum x^2}{n} - \frac{(\sum x)^2}{n}} = \frac{\sum xy - \frac{\sum x.\sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

Normal equation (Linear Model)

If the linear equation $w = a + bt$ then the corresponding normal equation becomes

$$\sum w = na + b\sum t$$

$$\sum wt = a\sum t + b\sum t^2$$

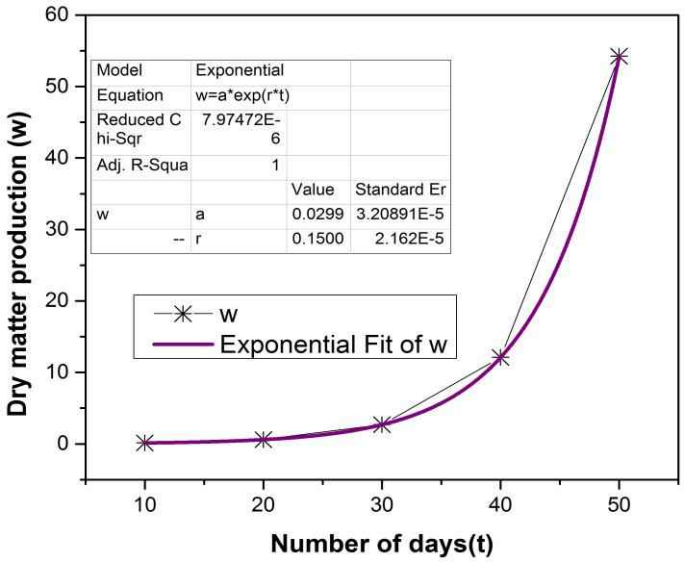
2) Exponential Model:

The Exponential growth Model will be of the form $W = A\,e^{rt}$. Here 'W' stands for the total dry weight of the plant at time't'. In this Model when t = 0, the value of W = A. Therefore, 'A' stands for the initial dry weight, i.e. The dry weight of the seed at the time of sowing. $W = A\,e^{rt}$

Example:

The following data gives the Green gram dry matter accumulation during different days. Calculate the seed dry weight and growth rate of Green gram using exponential growth model.

t in days	10	20	30	40	50
DMP W gms/plant	0.13	0.6	2.7	12.1	54.24



Example 2: Fit an exponential model to the following data.

t in days	5	15	25	35	45
W in mg per plant	0.05	0.4	2.97	21.93	162.06

Fitting up of Exponential Model

For fitting exponential model (**W = A e^{rt}**), the following steps are followed.

Step:-1

Take logarithmic transformation and convert into linear one.

$$\log w = \log\left(Ae^{rt}\right)$$

$$\log w = \log A + \log\left(e^{rt}\right)$$

$$\log w = \log A + rt$$

$$w_1 = a + bt$$

Assume $\log(w) = w_1$, $\log A = a$, $rt = bt$, $b = r$

Step:-2

Now apply methods of least squares principle and find out the constant a and b.

Step:-3

Finally take antilog of a, we can get the value of A. Then substitute the values of a and b in the exponential model.

The fitted exponential model is $\mathbf{W = A\, e^{rt}}$

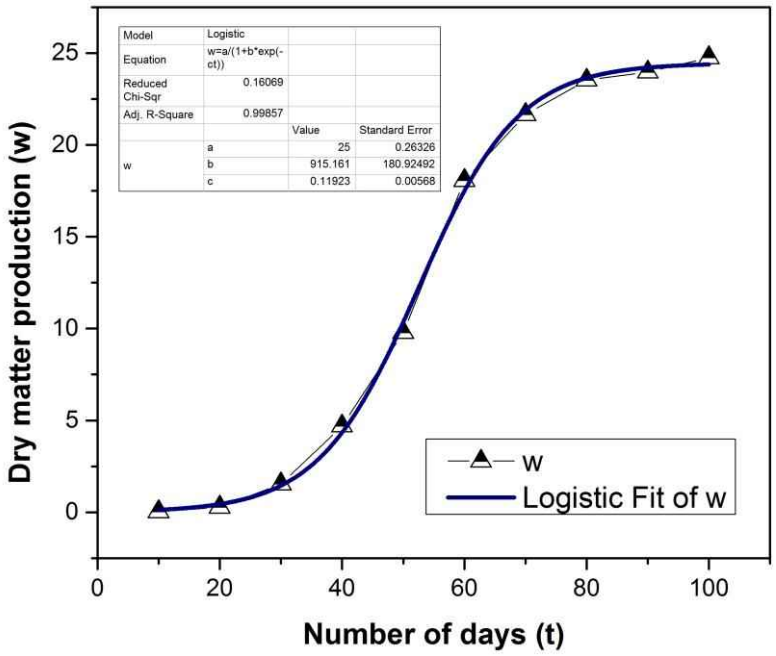
3) Logistic Model

The logistic Model will be of the form $W = \frac{a}{1+be^{-ct}}$

Example:

Consider the following data on dry matter accumulation in the buckwheat plants during vegetation.

Time (days)t	10	20	30	40	50	60	70	80	90	100
DMP	0.024	0.274	1.54	4.7	9.78	18.06	21.65	23.52	23.96	24.75



Fitting up of logistic model

For fitting logistic model ($W = \frac{a}{1+be^{-ct}}$) the following steps are followed.

Step:-1 Take reciprocals on both sides

$$\frac{1}{W} = \frac{1+be^{-ct}}{a} \Rightarrow \frac{a}{W} = 1+be^{-ct} \Rightarrow \frac{a}{W} - 1 = be^{-ct}$$

$$\Rightarrow \quad \frac{a-W}{W} = be^{-ct}$$

Step:-2

Take logarithmic transformation and convert into linear one.

$$\log\left(\frac{a-W}{W}\right) = \log\left(be^{-ct}\right)$$

$$\log\left(\frac{a-W}{W}\right) = \log b + \log\left(e^{-ct}\right)$$

$$\log\left(\frac{a-W}{W}\right) = \log b - ct$$

Assume $\log\left(\frac{a-W}{W}\right) = W;\ \log b = A; -ct = Bt$

Step:-3

Now apply methods of least squares principles and find out the constant A and B.

Step: - 4

Take antilog of A, we get the value of b and take –B get the value of c.

Finally substitute the values of a, b and c values in the model.

We get the fitted logistic model as $W = \frac{a}{1+be^{-ct}}$

Quadratic Model

The general form of a quadratic Model is $y = a + bx + cx^2$ where a, b and c are constants. Usually plant growth will not occur in the form of a Quadratic Curve. But in crop response curves, cost curves, supply curves and demand curves can be in this form.

Normal equation (Quadratic Model)

The general form of a quadratic model is $y = a + bx + cx^2$ then the corresponding normal equation becomes

$$\sum y = na + b\sum x + c\sum x^2$$

$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$

$$\sum x^2 y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

Response Model

The response Model between the input x and output y is of the form

$$y = a + bx - cx^2$$

UNIT I (Permutation and Combination, AP, GP, HP)

1. Permutation means ***Arrangement***
2. An arrangement of finite numbers of objects taken some or all at a time is called their ***Permutation***
3. Combination mean ***Selection***
4. The value of ${}^6C_1$ is ***6***
5. The value of 6P6 ***6!***
6. The value of nC0 ***1***
7. The trace of the matrix is ***sum of the leading diagonal elements***
8. The value of ${}^{n+1}C_r = {}^nC_r + {}^nC_{r-1}$
9. 6P4 is equal to ***360***
10. There are 30 people in a group. If all shake hands with one another, how many handshakes are possible? **435 (30C2)**
11. Relationship between ${}^nC_r$ and ${}^nP_r$ is ${}^nC_r = \frac{{}^nP_r}{r!}$
12. Number of permutations of 'n' things, taken 'r' at a time, when a particular thing is to be always included in each arrangement ***r * (n-1)P(r-1)***
13. If ${}^nC_{12} = {}^nC_8$, then n is equal to ***n=20 (nCr=nCn-r, r=12, n-r=8, n-12=8, n=20)***
14. A box contains 4 red, 3 white and 2 blue balls. Three balls are drawn at random. Find out the number of ways of selecting the balls of different colors ***24***
15. . In how many ways can a team of 3 boys and 3 girls be selected from 5 boys and 4 girls? ***40***
16. ${}^nC_r = {}^nC_{n-r}$ if $n > \frac{r}{2}$
17. ${}^nC_r + {}^nC_{r-1} =$ _____ ${}^{(n+1)}C_r$
18. The value of ${}^6C_5$ is equivalent to ${}^6C_1$
19. For a negative integer n, factorial n ***doesn't exist***
20. The number of permutation can be formed from the word MATHS is ***5!***
21. The number of total permutation letters of the word BANANA are ***60***
22. Letters SAP taken all at a time can be written as **6**

23. There are 6 varieties on brinjal, in how many ways these can be arranged in 6 plots which are in a line ***720***
24. From a group of 15 farmers in a village, a team of 11 farmers can be chosen in the ***1365*** ways
25. The number of permutation of n things taken r at a time, when m specified things always come together is ***m!(n-m+1)!***
26. In how many ways can selection of one cotton variety, one ragi variety be made from 10 cotton varieties and 5 ragi varieties ***50***
27. In how many waus can 5 persons be seated at a round table conference ***4!***
28. The value of ${}^{5}P_{2}$ =20
29. The order of matrix B=[1 2 5 7] is ***1x4***
30. A square matrix A is said to be non-singular if $|A| \neq 0$
31. If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ then A^{2013} 2^{2012} A
32. The transpose of cofactor matrix is the ***Adjoint matrix***
33. A square matrix in which all the elements above the leading diagonals are completely zero is called ***Lower triangular matrix***
34. A square matrix in which all the elements below the leading diagonals are completely zero is called ***Upper triangular matrix***
35. A square matrix A is symmetric if $a_{ij} = a_{ji}$
36. If two rows (columns) are proportional the determinant is equal to ***zero***
37. Find x, y, z from the matrix $\begin{bmatrix} 4 & 3 \\ x & 5 \end{bmatrix} = \begin{bmatrix} y & z \\ 1 & 5 \end{bmatrix}$ ***1, 4, 3***
38. If $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$ then x ***64***
39. The matrix $\begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$ is known as ***Column matrix***
40. The inverse matrix $A^{-1} = \frac{1}{|A|} \text{Adj } A$
41. The transpose of any matrix is obtained by interchange the ***rows and columns***
42. The rows are changed into columns then the determinant value is ***unaltered***
43. If A and B are two matrices then the product AB is possible iff the number of columns of matrix A is ***equal to number of rows of Matrix B***

44. A matrix P is order 2x3 and Q is of order 3x2. Then QP is of order ***3x3***

45. When A and B are singular matrices then AB is a ***Singular*** matrix

46. . If $A = -A^T$ then the matrix is called ***Skew Symmetric matrix***

47. Diagonal matrix in which all the leading diagonal elements are equal to one is known as ***Identity matrix***

48. If A is a square matrix of order 3 then $|kA|$ is $k^3|A|$

49. A matrix containing one column and several rows is called ***Column Matrix***

50. If any two row (columns) are interchanged then determinant is multiplied by ***-1***

51. All elements except the elements of the principal diagonal of a square matrix are zero, the matrix is called ***Diagonal Matrix***

52. A matrix whose elements are all equal to zero is known as ***Zero Matrix***

53. If the order of the matrix A is mxn then the order of A^T is ***nxm***

54. Two Matrices are equal iff both matrices are of ***Same*** type

55. If AB=BA =I then B is called ***Inverse of A***

56. The Hessian Matrix can be considered related to ***Jacobian Matrix***

57. The matrix the vertical arrangement are known as ***Column***

58. The inverse of the matrix A exists if $|A| \neq 0$

59. If $A = [a_{ij}]_{2x2}$ and $a_{ij} = i + j$ then $A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

60. A symmetric matric reproduces itself is known as ***Idempotent Matrix***

61. Two matrices A and B are added if ***they are of same order***

62. A scalar matrix having each element equal to 1 is said to be ***Unit matrix***

63. The value of ${}^7C_7$ is 1

64. A square matrix A is said to be symmetric if $A^T=A$

65. State Incorrect statement
    a) A^{-1} exists for non singular matrices
    b) Determinant of A and A^T are same
    c) Permutation means arrangement of things
    d) **From a group of 15 farmers in a village a team of 11 farmers can be chosen in ${}^{15}P_{11}$**

66. ${}^{10}C_3 = 120$

67. The matrix $\begin{pmatrix} 0 & 3 & -4 \\ -3 & 0 & 5 \\ 4 & -5 & 0 \end{pmatrix}$ is known as **skew symmetric**

68. If $(n+1)C_3 = 2(nC_2)$ then the value of n is **5**

69. The value of the product of $\begin{vmatrix} 1 & 2 \\ -3 & 1 \end{vmatrix} x \begin{vmatrix} 2 & 4 \\ 1 & -1 \end{vmatrix}$ is **-42**

70. The value of 8P3 is **336**

71. How many rectangular plots can be formed by joining the vertices of a site in the form of hexagon is **15**

72. The number of ways to arrange n distinct objects along a fixed circle is **n!**

Arithmetic Progression, Geometric Progression, Harmonic Progression

1. The 10th term of the series 1, 4, 7,… is **28**
2. Arithmetic Progression=Common difference

UNIT II (Straight line-different forms, circle)

1. The midpoint of the line joining $(a,-b)$ and $(3a,5b)$ is ***(2a, 2b)***
2. If $a_1b_2 - a_2b_1$ tends to zero both x_1 and y_1 tends to infinity and the lines will be parallel then **the point of intersection shifts to infinity**
3. The point P which divides the line segment joining the points $A(1,-3)$ and $B(-3,9)$, internally in the ratio 1:3 is ***(0,0)***
4. The angle of inclination of a straight line parallel to x-axis is equal to ***zero***
5. Slope of the line joining the points, $(3,-2)$ and $(-1,a)$ is $-\frac{3}{2}$ then the value of a is equal to ***4***
6. The straight line $4x+3y-12=0$ intersects the y- axis at ***(0,4)***
7. The slope of the straight line $7y-2x=11$ is equal to ***2/7***
8. The equation of a straight line having slope 3 and y-intercept -4 **is** ***y=3x-4***
9. The x and y-intercepts of the line $2x-3y+6=0$, respectively are ***-3, 2***
10. The shortest distance between any two points is a ***Straight line***
11. If B divides A(3, 2) and B(1,0) externally in the ration 2:3, what are coordinates of B ***(7,6)***

12. Find the equation of the straight line passing through the points (0,15) and (3, 0)

    5x+y-15=0

13. Find the distance between A (4, –3) and B (–2, 5) ***10***

14. Equation of straight line with slope -8 and y intercept (0,16) is written as ***y=-8x+16***

15. Find the distance of the point (–6, 8) from the origin ***10***

16. If the distance between the points (4, c) and (1,0) is 5, then c is equal to ***4***

17. The end points of diameter of circle are (2, 4) and (-3, -1). The radius of the circle is

    $5\sqrt{2}/2$

18.

19. The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a circle, the condition will be ***a=b and h=0***

20. The equation of the line having intercept on the axes such that the sum of the intercepts is 7 and the product of the intercepts is 12 is $\frac{x}{4} + \frac{y}{3} = 1$

21. A (4, 7) and B (7, 3) then AB is equal to ***5***

22. If a straight line makes an angle θ with the positive direction of the x-axis, then θ is known as ***angle of inclination***

23. If a straight line makes an angle θ with the positive direction of the x-axis, then θ is known as ***Slope***

24. If the origin is shifted to (1, 2, –3) without changing the direction of the axis then find the new coordinates of the point (0, 4, 5) with respect to the new frame is **(-1, 2, 8).**

    x' = x – x1, where (x1, y1, z1) is the shifted origin
    y' = y – y1
    z' = z – z1
    x' = 0 – 1 = –1
    y' = 4 – 2 = 2
    z' = 5 + 3 = 8

25. Find the equation of the straight line which cuts off an intercept of 5 units from y axis and makes with the x-axis an angle of 60° $y = \sqrt{3}x + 5$

26. Find the equation of the straight line passing through (2, 3) and making an angle of 30° with x-axis $y - 3 = \frac{1}{\sqrt{3}}(x - 2)$

27. The path traced out by a moving point under the given condition is called as ***Locus***

28. The equation of the line which is parallel to the straight line 3x + 4y -7= 0 and passing through (1, 2) is ***3x+4y-11=0***

29. The diameter of the circle $x^2 + y^2 - 4x - 6y = 0$ is $\sqrt{52}$
30. The point of intersection of the lines 3x + 4y = 0 and 5x – 6y = 0 is ***(0,0)***
31. Determine the equation of the line whose perpendicular distance from the origin is 8 units and makes an angle of 90° in positive direction of x-axis is ***y=8***
32. Find the length of the perpendicular from (3, 2) on the line 3x+2y+1=0 $\frac{14}{\sqrt{13}}$
33. Find the length of the perpendicular from P(1, 3) to the line 2x-3y+10=0 is $\frac{3}{\sqrt{13}}$
34. What is the slope of the line $y = -\frac{2}{3}x + \frac{3}{4}$
35. The two lines ax + by = c and a′x + b′y = c′ are perpendicular if $aa' + bb' = 0$
36. Find the angle between the lines 2x + y – 4 = 0 and 3x – y + 1 = 0 is 45^0
37. When two or more lines are parallel then their slopes are ***equal***
38. If the angle between the two lines is zero then the lines are ***Parallel***
39. When two lines are perpendicular then the product of their slopes is ***-1***
40. Find the equation of the line 3x+4y+28=0 perpendicular to the line through the point (-1, 4) is ***4x-3y+16=0***
41. Find the equation line passing through (-3, 5) and perpendicular to the line through the points (2, 5) and (-3, 6) ***5x-y+20=0***
42. Slope of any line parallel to x-axis is ***zero***
43. Slope of any line parallel to y-axis is ***infinite***
44. Find the Centre and radius of the circle $x^2 + y^2 - 6x + 4y + 2 = 0$ is ***(3, -2) and*** $\sqrt{11}$
45. Find the equation of the circle whose centre is (5,-7) and passing through the point (3,-3) $x^2 + y^2 - 10x + 14y + 54 = 0$
46. When three or more lines intersect at the same point is known as ***Concurrent***
47. Circle have same Centre and different radii is called ***Concentric*** circles
48. Find the equation of a circle that has points P (1, 8) and Q (5, -6) as the endpoints of a diameter $x^2 + y^2 - 6x - 2y - 43 = 0$
49. Find the length of the perpendicular to the line 4x+6y+7=0 from the origin $\frac{7}{\sqrt{52}}$
50. Determine the equation of the line with x-intercept 2 and y-intercept 3. ***3x+2y=6***
51. Find the center of the circle having the following equation $x^2 + y^2 - 6x + 8y - 24 = 0$ ***(3, -4)***

52. Find the slope of the line y=3x ***3***

53. A point on x-axis which equidistance from the points (3, 4) and (2,5) is ***(-2, 0)***

54. Write the equation of the line passing through (2, 3) and parallel to x-axis **y=3**

55. The x-coordinate of every point on the y-axis is *zero*

56. What is the slope of a line which makes an angle of 45^0 with the positive direction of y axis is ***-1***

    (y-axis=90+yaxis degree=135, tan(135)=-1)

57. If a circle does not pass through origin and the center lies on the x-axis then

    $(x-a)^2 + y^2 = r^2$

58. The perpendicular distance of the origin (0, 0) from Ax+By+C=0 is $\frac{C}{\sqrt{A^2+B^2}}$

59. State correct statement

    a) The y intercept of the straight line x+3y-5=0 is 3/5

    b) The slope of the straight line 2x+y-4=0 is 2

    c) The straight lines are perpendicular then the product of their slopes equal to one

    d) The equation of the line passing through (1,2) and perpendicular to x+y+7=0 is y-x-1=0

60. State **incorrect** statement

    a) Two or more circles having same centre are called concentric circles

    b) The equation of the circle with radius r and center at origin is $x^2 - y^2 = r^2$

    c) The equation of the straight line parallel to ax+by+c=0 is ax+by+k=0

    d) The midpoint of the line joining (5,6) and (7,6) is (6,6)

61. Find the distance between A (1, 2) and B (3, 2) is **2**

62. Find the equation of the straight line passing through the points (-5,-23) and (-2, -8) is **y=5x-2**

63. If the equation of the straight line is $y = \sqrt{3}x + 4$, then the angle made by the straight line with the positive direction of x –axis is $\mathbf{60^0}$

UNIT III (Limit and continuity, Increasing and decreasing function, Differentiation, Partial Differentiation)

1. The value of $\lim_{x \to a} \frac{x^n - a^n}{x - a}$ is na^{n-1}

2. The value of $\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$

3. The value of $\lim_{x \to a} \frac{x^{5/8} - a^{5/8}}{x^{1/3} - a^{1/3}}$ is $\frac{15}{8} a^{7/24}$
4. A function y=f(x) can have more than maximum or minimum point. The maximum of all maximum points is called **Global maximum**
5. For the following function obtain update level of fertilizer application for which yield in maximum and find the maximum yield $y = 1800 + 7x - 0.03x^2$ is **2208.34**
6. If the cost function is $C=2Q+3Q^2$ then the marginal cost function is **2+6Q**
7. The necessary conditions for physical optimum is $\frac{dy}{dx} = 0$
8. The condition for the function y=f(x) to be increasing is $\frac{dy}{dx} > 0$
9. D(logx)=1/x
10. The function y=4-6x is always **decreasing function**
11. Maxima and minima of several variable- **Hessian matrix**
12. The minimum of all the minimum point is called **Global Minimum**
13. Maximum and minimum points are also called **stationary** points
14. The necessary condition for Economic Optimum is $\frac{dy}{dx} = \frac{p_x}{p_y}$
15. Neither maximum nor minimum such a point is called **Saddle point**
16. The marginal cost is denoted by $MC = \frac{dc}{dq}$
17. The function $x^3 + 3x^2y + xy^2 - y^3 = 0$ is homogeneous function of degree **3**
18. The function $y = x^3 + 2$ is **increasing** for all values of x
19. The formula for relative growth rate is $RGR = \frac{1}{W}\frac{dw}{dt}$
20. Identify the **incorrect** statement
    a) Differentiation of e^x with respect to x is e^x
    b) Differentiation of cost function gives marginal cost
    c) Y is increasing if $\frac{dy}{dx} > 0$
    d) Y is increasing if $\frac{dy}{dx} < 0$
21. Identify the **correct** statement
    a) The process of finding the co efficient is called differentiation
    b) The process of finding the limits is called differentiation
    c) The process of finding the derivative is called integration
    d) The process of finding the derivative is called differentiation
22. Identify the **incorrect** statement
    a) The differential coefficient of e^x is logx
    b) The differential coefficient of x^n is nx^{n-1}
    c) The differential coefficient of 5 is 0

d) The differential coefficient of a constant c is 0

23. Identify the **correct** statement

a) The first order condition for y being a maximum or a minimum is $\frac{dy}{dx} \neq 0$

b) The first order condition for y being a maximum or a minimum is $\frac{dy}{dx} < 0$

c) The first order condition for y being a maximum or a minimum is $\frac{dy}{dx} = 0$

d) The first order condition for y being a maximum or a minimum is $\frac{dy}{dx} > 0$

24. $\frac{dc}{dx}$ is **Marginal cost**

25. The rate of change of y with respect to x is denoted as $\frac{dy}{dx}$

26. Maximum function $\frac{d^2y}{dx^2} < 0$

27. Decreasing function $\frac{dy}{dx} < 0$

28. $\frac{d}{dx}(8)$ is **zero**

29. $\frac{d}{dx}(\sec x)$ is secxtanx

30. If R is the revenue function for x units, $\frac{dR}{dx}$ gives **Marginal Revenue**

31. The function $x^3 - 2x^2y + y^3 = 0$ is homogeneous function of degree **3**

32. The necessary condition for f(x) being maximum or a minimum at x=a is $f'(x) = 0$

33. The growth rate of $Y = t^2 + 3t + 4$ is **2t+3**

34. The derivative of the function y=log(4x) is **1/x**

35. $\frac{d}{dx}(e^x)$ is e^x

36. Identify the **correct** statement

a. Differentiation of $2x^2$ with respect to x is 2

b. For any positive integer n , $\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}$

c. The derivative of secx is with respect to x is cosecx

d. A function of the form $f(x, y) = 0$ **is called an implicit function**

37. Identify the incorrect statement

a) Differentiation of a constant is one

b) For any positive integer n , $\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}$

c) The derivative of tanx with respect to x is $\sec^2 x$

d) $\lim_{x \to a} \frac{\sin x}{x} = 1$

38. Minimum function $\frac{d^2y}{dx^2} > 0$

39. $\frac{d}{dx}\left(e^{6x}\right)$ is $6e^{6x}$

40. Average revenue of the revenue function $R = 3Q^3 + 6Q$ is $9Q^2 + 6$

41. If $\frac{dg}{dt} = 7t + t^2$ is the growth rate then the growth function is $7\frac{t^2}{2} + \frac{t^3}{3}$

42. $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = nu$ is known as **Euler's theorem**

43. The derivative of a^x is $a^x \log a$

44. The function in which each term has same degree is called **homogeneous function**

45. For what value of x the function $y = x^2 - 3$ is increasing **x>0**

46. **If** $x = b(1 + \cos\theta)$**,** $x = b(\theta + \sin\theta)$ **then** $\frac{dy}{dx} = \frac{1 + \cos\theta}{-\sin\theta}$

47. A function y=f(x) is exist $f(x_1) < f(x_2)$ for all $x_1 < x_2$ then it is called **increasing function**

48. Let x-time, y-fertilizer, then w=f(x,y) dw/dx is called **absolute growth rate**

49. The reciprocal of cosx is **secx**

50. The necessary condition for the function y=f(x) to have a maximum is $\frac{dy}{dx} = 0$

UNIT 1V (Integration)

1. $\int \sec x \tan x dx$ is **secx+c**

2. $\int \sin x dx$ is **–cosx+c**

3. $\int \sin^2 x dx$ is $\frac{1}{2}\left(x - \frac{\sin 2x}{2}\right) + c$

4. The value of $\int \frac{dx}{x^2 - 4x + 8}$ **is** $\frac{1}{2}\tan^{-1}\left(\frac{x-2}{2}\right) + c$

5. $\int \frac{dx}{1 + x^2} = \tan^{-1} x + c$

6. $\int \frac{dx}{x^2 - 4} = \frac{1}{2}\log\left(\frac{x-2}{x+2}\right) + c$

7. $\int \sin x \cos x dx = \frac{-\cos 2x}{4} + c$

8. The value of $\int \operatorname{cosec} x \cot x dx = \frac{-1}{\sin x} + c$

9. The value of $\int \frac{1}{\sqrt{3+x^2}} dx = \log(x + \sqrt{x^2+3}) + c$

10. The value of $\int \frac{dx}{x}$ is **logx**

11. The value of $\int_{-3}^{3} xdx$ is __

12. $\int_{0}^{\frac{\pi}{2}} \sin x dx$ **is 1**

13. State **incorrect** statement
    a) Integration is the inverse process of differentiation
    b) $\int \cos x dx = \sin x + c$
    c) $\int \frac{dx}{1+x^2} = \tan^{-1} x + c$
    d) $\int \frac{2xdx}{1+x^2} = \tan^{-1} x + c$ **is $\log(1-x^2)+c$**

14. Identify **correct** statement
    a) $\int udv = uv + \int vdu$
    b) $\int \cos x dx = -\sin x + c$
    c) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + c$ ******
    d) $\int \log x dx = \frac{1}{x} + c$

15. The value of $\int_{0}^{2} x^3 dx$ **is 4**

16. $\int 5 dx$ **is 5x**

17. $\int_{0}^{2} xdx$ **is 2**

18. $\int \tan x dx$ **is log(secx)**

19. Identify the **incorrect** statement
    a) Integration is not the inverse process of differentiation
    b) Differentiation of total revenue function gives marginal revenue function
    c) Integration of marginal revenue function gives total revenue function
    d) Integration is the inverse process of differentiation

20. Identify the incorrect statement
    a) The Integration of x^2 is $\frac{x^3}{3}$

b) The integration of x is $\frac{x^2}{2}$

c) The integration of sinx is –cosx

d) The Integration of x^2 is $\frac{1}{x}$

21. The area under the curve y=f(x), the x-axis and the coordinates at x=a and x=b is $\int_a^b ydx$

22. The value of $\int_{-3}^{3} xdx$ is ---

23. Integration by parts formula is $\int udv = uv - \int vdu$

24. $\int x^5 dx = \frac{x^6}{6} + c$

25. The value of $\int 5^x dx$ **is** $\frac{5^x}{\log 5} + c$

26. Area included between the two curves is $\int_a^b (f(x) - g(x))\,dx$

27. The area bounded by y=x, y axis y=0 and y=1 is $\int_0^1 ydy = \frac{1}{2}$

28. If the marginal cost function is MC is= $= 3e^{3x}$, then the cost function is e^{3x}

29. Integration of e^{7x} is $e^{7x}/7$

30. The value of $\frac{1}{\sqrt{x^2 - a^2}}dx$ is $\log\left(x + \sqrt{x^2 - a^2}\right)$

31. The general form of integration by parts is **Bernoullis formula**

32. The value of $\int_0^1 xe^x dx$ is 1

33. $\int \log x dx = (x \log x - x) + c$

UNIT V (Mathematical Modeling)

1. Identify the **correct** statement
   a) W=a+bt is a linear model
   b) W=a+bt is a exponential model
   c) W=a+bt is a logistic model
   d) W=a+bt is a abstract model
2. Identify the **correct** statement
   a) In the logistic model ‘b’ is known as the carrying cost
   b) In the logistic model ‘a’ is known as the carrying cost
   c) In the logistic model ‘t’ is known as the carrying cost
   d) In the logistic model ‘w’ is known as the carrying cost
3. State **incorrect** statement
   a) Model of a dam is a physical model
   b) A system is represented by a set of equations, then it is called mathematical model

c) In linear equation, the parameter 'a' indicates seed weight

d) Quadratic model is a linear model

4. State **correct** statement

a) A model is defined as a physical representation of any natural phnenomena

b) W=bt is a linear model with constant term

c) In a linear model is y=a+bx the variables x and y are degree 2

d) Growth rate of an exponential model is $w = ae^{bt}$ is b

5. Let $(x_1 - y_1), (x_2 - y_2), ..., (x_n - y_n)$ be n pairs of observation. Then plotting these points on an ordinary graph sheet , We get a collection of dots called **Scatter diagram**
6. $w = ae^{bt}$ is a **Exponential Model**
7. A linear model without a constant term is **w=bt**
8. The crop response model is also known as **quadratic model**
9. The quadratic model is **Nonlinear Model**
10. $y = a + bx + cx^2$ is **quadratic model**
11. The relative growth rate of $w = ae^{bt}$ is **b**
12. The linear model is of the form is **w=a+bt**
13. Principle of least squares **sum of the squares of the residuals must be minimum**
14. The general form of quadratic model is $y = a + bx + cx^2$
15. Empirical models are also called **Functional** Models
16. Malthusian parameter is related to **Exponential Model**
17. Intensive growth rate is known as **Malthusian parameter**
18. The mathematical equation of exponential model is $w = ae^{bt}$ if t=0 then **w=a**
16. The general form of a Linear Model is w=a+bt. It is daid to be linear since the highest degree for w and t is **one**
19. The logistic model is of the form $w = \frac{a}{1 + be^{-ct}}$
20. Demand curve is a curve of **Sloping downwards curve**
21. $y = ax^2 + b$ is a **Quadratic model**
22. $y = at^2 + bt$ is a **Quadratic model**
23. Miniature building model is a **Physical Model**